Unit

31 Typical reactions of selected functional groups

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Typical reactions of alkanes, alkenes, haloalkanes, alcohols, aldehydes, ketones, carboxylic acids, esters and amides

31.1 Introduction

The synthesis of carbon compounds is very important in the research and production of new and useful products.

Suppose we need to plan a synthetic route to accomplish the following conversion:

$$CH_3CH_2CHO \xrightarrow{?} CH_3CH{=}CH_2$$

The synthesis may take two steps as follows:

$$CH_3CH_2CHO \xrightarrow{\text{reduction}} CH_3CH_2CH_2OH \xrightarrow{\text{dehydration}} CH_3CH{=}CH_2$$

To be able to design such a synthesis, we require a good knowledge of typical reactions of the functional groups concerned (i.e. $-\overset{\overset{O}{\|}}{C}-H$ and $-OH$) and the conversion of one functional group into another.

In this unit, we are going to discuss important reactions of members of some homologous series.

31.2 Important reactions of alkanes

We have discussed two important reactions of alkanes in Topic 7 Fossil Fuels and Carbon Compounds. These are **combustion** and **substitution reactions**.

Combustion of alkanes

In a good supply of oxygen, alkanes undergo complete combustion to produce carbon dioxide and water, and release much heat. The following equation represents the complete combustion of methane, CH_4.

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$

Reaction with halogens — substitution reactions

The reaction of an alkane with chlorine occurs when a mixture of the two is irradiated with ultraviolet light or heated. For example, methane reacts with chlorine to yield a mixture of carbon compounds. The reaction of an alkane with a halogen is a substitution reaction, called **halogenation**.

$$CH_4(g) + Cl_2(g) \xrightarrow[\text{or heat}]{\text{UV light}} CH_3Cl(g) + HCl(g)$$

$$CH_3Cl(g) + Cl_2 \xrightarrow[\text{or heat}]{\text{UV light}} CH_2Cl_2(l) + HCl(g)$$

$$CH_2Cl_2(l) + Cl_2 \xrightarrow[\text{or heat}]{\text{UV light}} CHCl_3(l) + HCl(g)$$

$$CHCl_3(l) + Cl_2 \xrightarrow[\text{or heat}]{\text{UV light}} CCl_4(l) + HCl(g)$$

Reference Website

Refer to the following website for an animation showing the substitution reaction between methane and bromine:
http://resources.hkedcity.net/resource_detail.php?rid=143271527
(accessed July 2014)

Alkanes react with bromine in a similar manner.

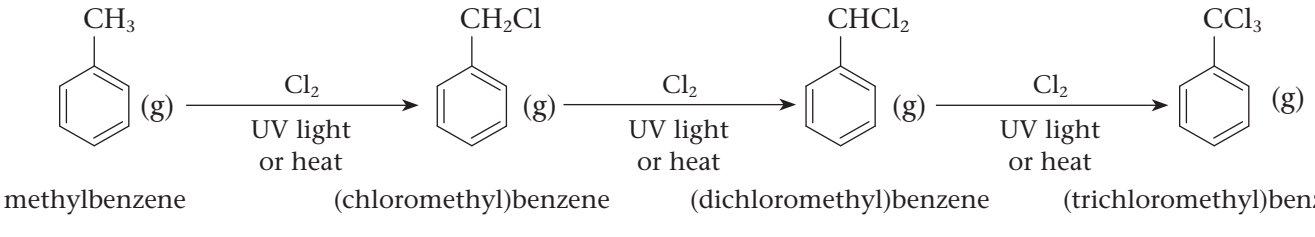
The products of the reactions between alkanes and halogens are known as haloalkanes.

CE 2008 Paper 2 Q24

combustion 燃燒　substitution reaction 取代反應　halogenation 鹵化作用

N1 The major product is (chloromethyl)benzene when using excess methylbenzene. This is NOT an effective way for the preparation of (chloromethyl)benzene because the reaction gives a mixture of products.

Chlorination of methylbenzene takes place in the gas phase in the presence of ultraviolet light or when heated. N1



methylbenzene (chloromethyl)benzene (dichloromethyl)benzene (trichloromethyl)benzene

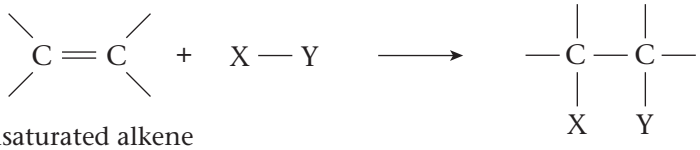
ASL 2011 Paper 2 Q7(a)
AL 2011 Paper 2 Q9(a)(i)

AL 2007 Paper 2 Q6(a)(ii)

31.3 Addition reactions of alkenes

In Topic 7, we learnt that the most characteristic reaction of compounds containing carbon-carbon double bonds is an **addition reaction**.

The general reaction is:



unsaturated alkene

a saturated compound

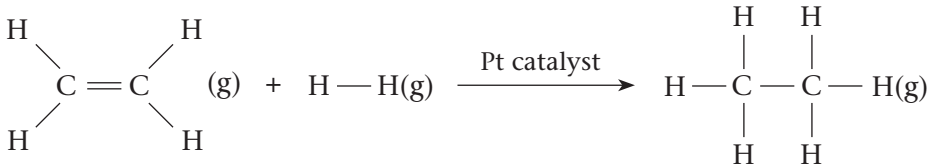
Addition of hydrogen to alkenes

In the presence of a catalyst, hydrogen adds across a carbon-carbon double bond of ethene to form ethane, which is a saturated compound.

If a platinum or palladium catalyst is used, the process takes place under normal laboratory conditions. Nickel is a cheaper but less efficient catalyst. It needs to be finely powdered and the gases need to be at 150 °C and 5 atmospheres for **hydrogenation** to occur.

ASL 2008 Paper 2 Q1(b)
AL 2008 Paper 2 Q7(a)

For example,



ethene ethane

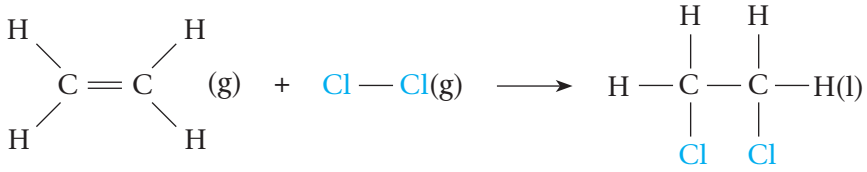
addition reaction 加成反應 hydrogenation 加氫作用

Addition of halogens to alkenes

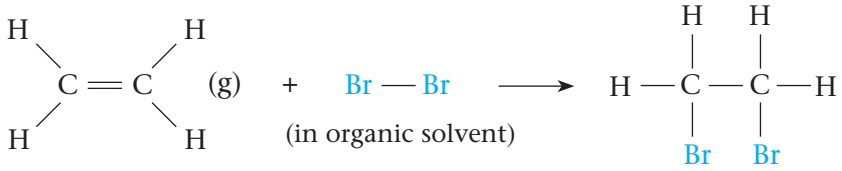
CE 2006 Paper 2 Q16
ASL 2008 Paper 2 Q1(b)
AL 2008 Paper 2 Q7(a)

Alkenes react quickly with chlorine gas, or bromine dissolved in an organic solvent, to give dihaloalkanes.

For example,



ethene 1,2-dichloroethane



ethene 1,2-dibromoethane (in organic solvent)

Ethene reacts explosively with fluorine, but the reaction with iodine is rather slow.

Test for unsaturation

CE 2009 Paper 2 Q39
ASL 2008 Paper 1 Q9(b)

Often the test for an alkene involves shaking the substance being tested with aqueous bromine rather than pure bromine. Decolourization of the aqueous bromine occurs quickly if an alkene is present (Fig. 31.1).

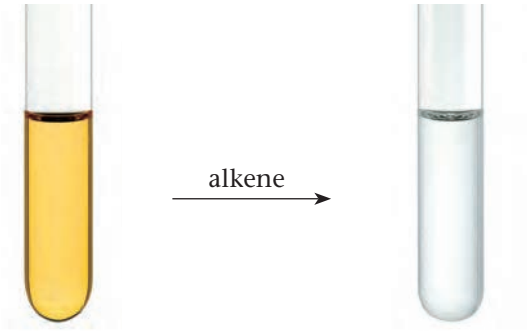
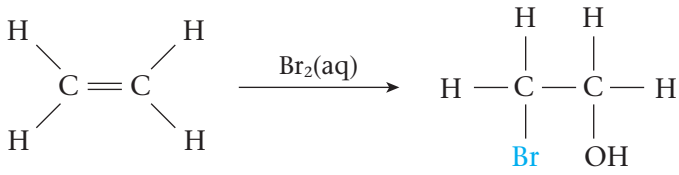


Fig. 31.1 Decolourization of aqueous bromine by an alkene

In the case of ethene with aqueous bromine, 2-bromoethanol is obtained.



2-bromoethanol

CE 2007 Paper 2 Q27
CE 2008 Paper 2 Q13

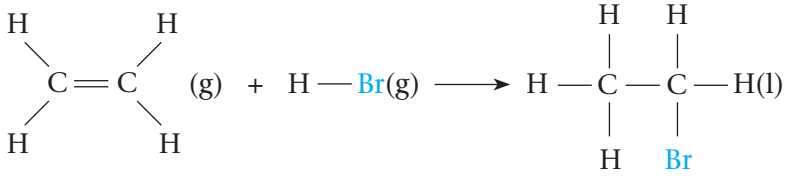
Remember that we can also use the addition reaction of an alkene with cold acidified dilute potassium permanganate solution as a test for alkenes. The permanganate solution changes from purple to colourless quickly when shaken with an alkene.

Addition of hydrogen halides to alkenes

CE 2005 Paper 2 Q4
ASL 2008 Paper 2 Q1(b)
AL 2008 Paper 2 Q7(a)
AL 2009 Paper 2 Q4(b)(ii)

Ethene reacts with gaseous hydrogen halides at room temperature.

For example,



bromoethane

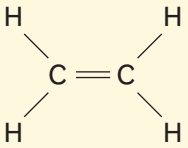
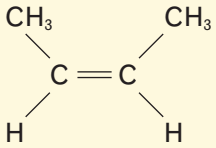
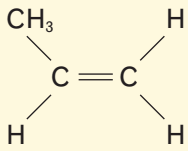
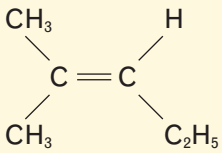
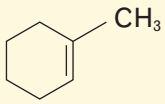
Ethene also reacts with concentrated aqueous solutions of hydrogen halides in the cold.

Addition of hydrogen halides to asymmetrical alkenes — Markovnikov's rule

An asymmetrical alkene, such as propene, is an alkene that has different groups or atoms attached to the carbon atoms of the C=C bond. Table 31.1 shows the structural formulae of some symmetrical and asymmetrical alkenes.

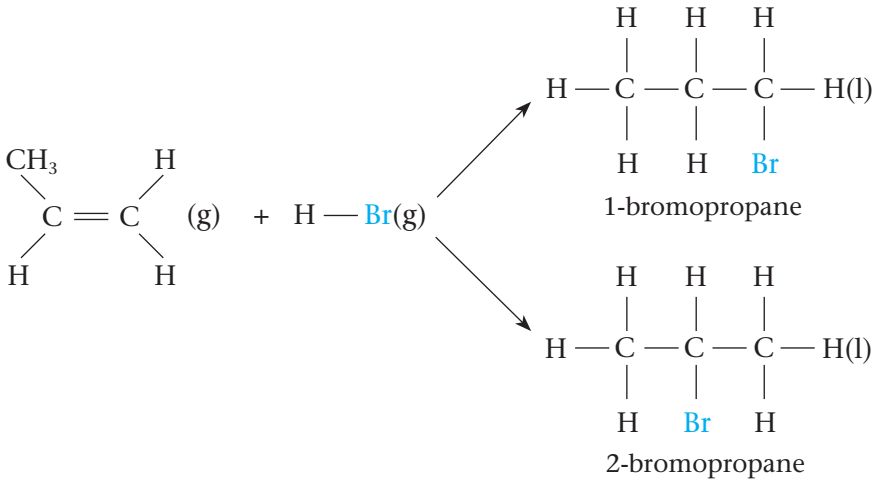
Table 31.1

Structural formulae of some symmetrical and asymmetrical alkenes

Symmetrical alkenes	 ethene	 but-2-ene	 cyclohexene
Asymmetrical alkenes	 propene	 2-methylpent-2-ene	 1-methylcyclohexene

When a hydrogen halide adds to an asymmetrical alkene, two products are possible.

For example, the addition of hydrogen bromide to propene could give either 1-bromopropane or 2-bromopropane.

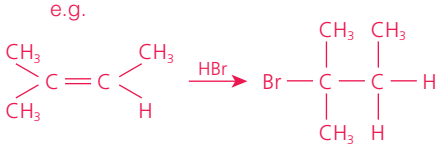


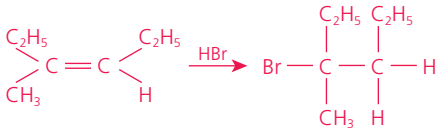
Markovnikov's rule 馬科尼科夫規則

N2 Predicting the major product using Markovnikov's rule

Examination questions may ask students to predict the major product of an addition reaction involving an asymmetrical alkene.

e.g.



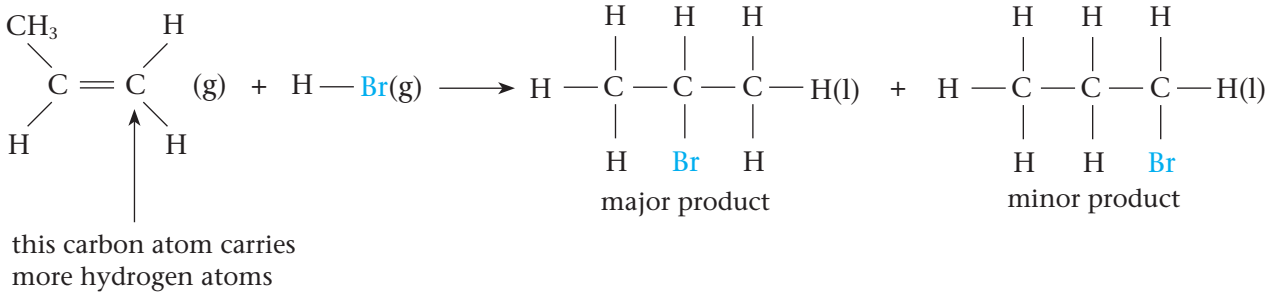


In fact, the product is almost entirely 2-bromopropane. The Russian chemist, Markovnikov, formulated a rule for predicting the major addition product formed. Markovnikov's rule says that:

> ✓ When a molecule HA adds to an asymmetric alkene, the major product is the one in which the hydrogen atom attaches itself to the carbon atom already carrying the larger number of hydrogen atoms. N2

ASL 2012 Paper 2 Q8(c)(i)
AL 2007 Paper 1 Q5(a)(ii)

Thus, for the reaction between propene and hydrogen bromide, 2-bromopropane is the major product.



For the reaction between 1-methylcyclohexene and hydrogen bromide, 1-bromo-1-methylcyclohexane is the major product.

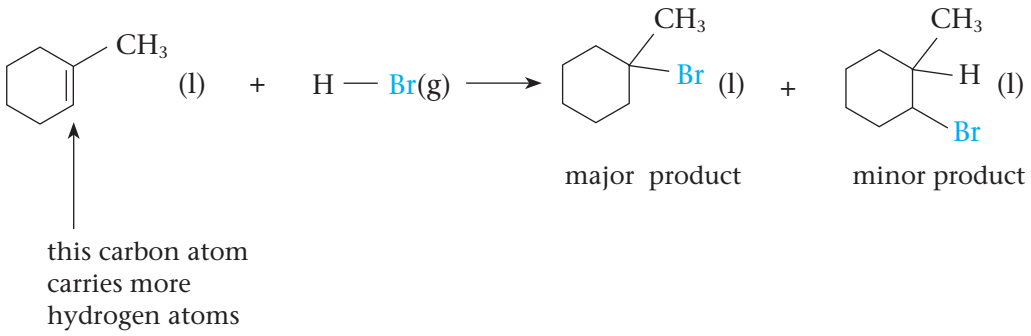


Fig. 31.2 summarizes the addition reactions of propene.

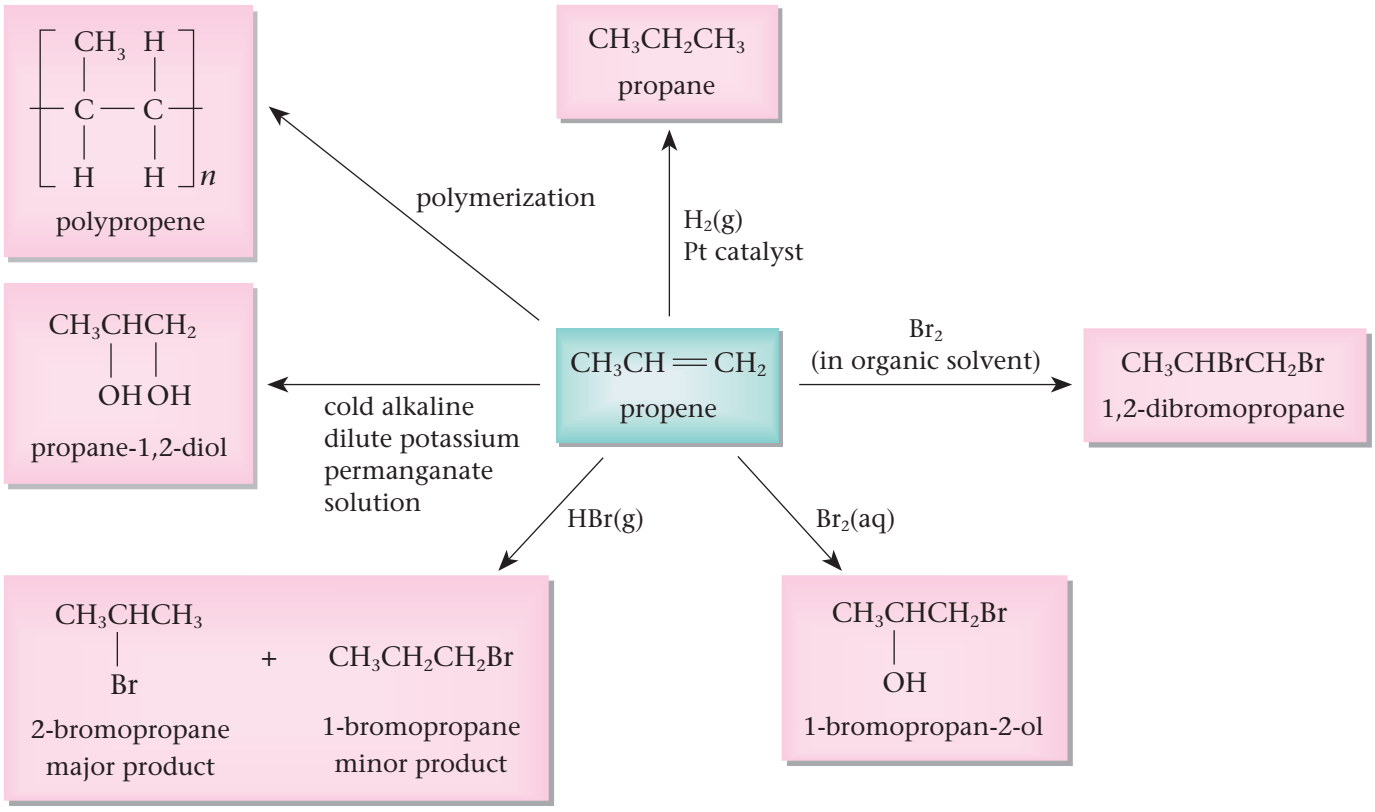
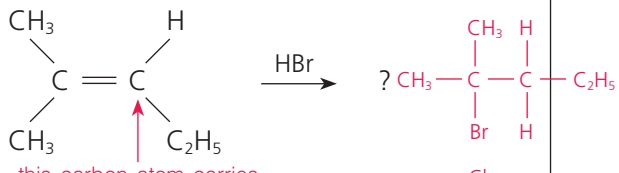


Fig. 31.2 A summary of the addition reactions of propene

Practice 31.1

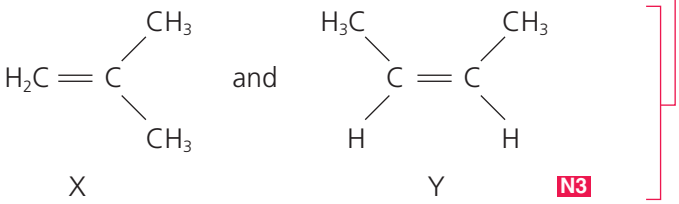
N3 Both X and Y react with H_2(g) in the presence of Ni(s). X and Y undergo polymerization to give different addition polymers. DSE 2013 Paper 1A Q31

1 Predict the major product of each of the following reactions.

a) 

b) $CH_3CH_2CH_2CH=CH_2 \xrightarrow{HCl}$? $CH_3CH_2CH_2CHCH_3$ (Cl)

2 Consider the isomeric compounds X and Y shown below:



a) Name the type of isomerism involved.
Structural isomerism

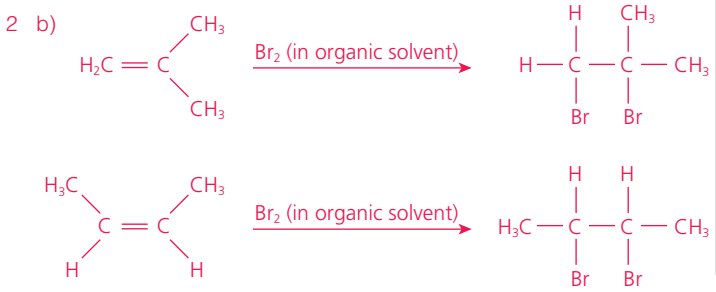
b) Both X and Y react with bromine dissolved in an organic solvent. Draw the structure of the carbon compound formed in each case.

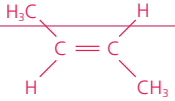
c) Y has a stereoisomer.

i) Name the type of stereoisomerism shown by Y.
Cis-trans isomerism

ii) Draw the structure of the stereoisomer of Y.

2 b)





31.4 Substitution reactions of haloalkanes

Substitution reactions are typical reactions of haloalkanes. For example, when a haloalkane is heated under reflux with sodium hydroxide solution, the halogen is substituted by an –OH group and an alcohol is produced. This reaction is sometimes called **hydrolysis** ('splitting with water').

$$\underset{\text{haloalkane}}{R{-}X} \xrightarrow[\text{reflux}]{NaOH(aq)} \underset{\text{alcohol}}{R{-}OH}$$

For example,

AL 2009 Paper 2 Q4(b)(ii)

$$\underset{\text{1-bromopropane}}{CH_3CH_2CH_2Br(l)} + OH^-(aq) \longrightarrow \underset{\text{propan-1-ol}}{CH_3CH_2CH_2OH(aq)} + Br^-(aq)$$

Fig. 31.3 shows the apparatus used for heating a haloalkane with sodium hydroxide solution.

The boiling points of haloalkanes are not high. We have discussed this in Unit 29.

Often ethanol is added to act as a solvent. In the reflux condenser, all the vapour is condensed back into the flask. This prevents any loss of the reaction mixture by evaporation.

If the reactant(s) or solvent used has a high boiling point (>130 °C), an air condenser should be used instead of a water condenser in reflux. This is because the large temperature difference outside and inside the water jacket may cause the water condenser to crack.

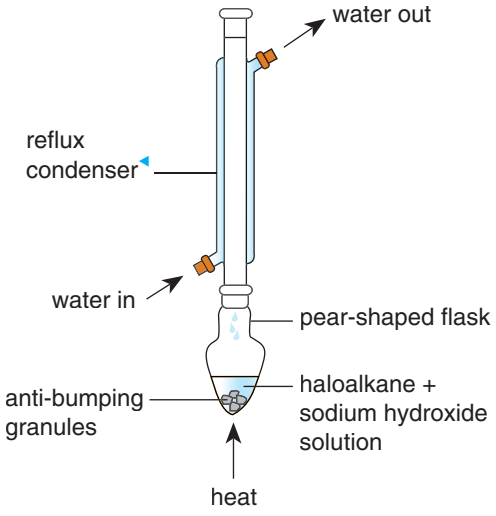


Fig. 31.3 Apparatus used for heating a haloalkane with sodium hydroxide solution under reflux

N4 Refer to Teacher's Resources Book page 34.

N5 Refer to Teacher's Resources Book page 35.

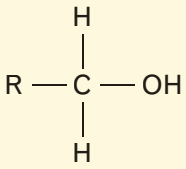
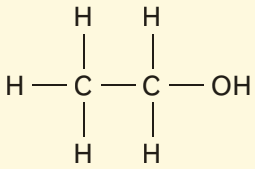
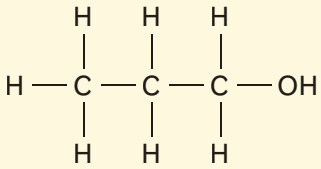
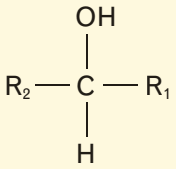
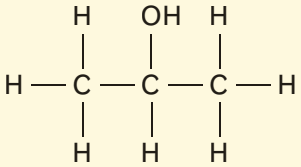
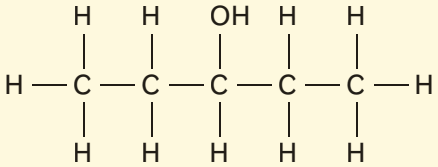
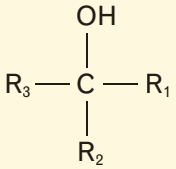
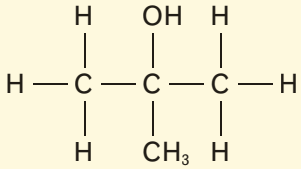
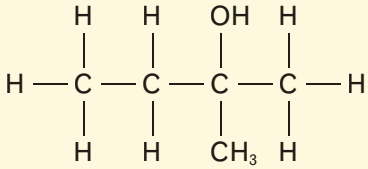
31.5 Reactions of alcohols N4 N5

There are three main categories of alcohols — primary (1°), secondary (2°) and tertiary (3°). This classification is based on the number of alkyl groups attached to the carbon atom bearing the hydroxyl group (–OH). A primary alcohol has one alkyl group attached, a secondary alcohol has two alkyl groups attached and a tertiary alcohol has three alkyl groups attached (Table 31.2).

Methanol (CH_3OH) is classified as a primary alcohol.

Table 31.2

General structures and examples of different categories of alcohols

	General structure	Examples
Primary (1°) alcohol		 ethanol  propan-1-ol
Secondary (2°) alcohol		 propan-2-ol  pentan-3-ol
Tertiary (3°) alcohol		 methylpropan-2-ol  2-methylbutan-2-ol

Although all alcohols behave similarly in some reactions, primary, secondary and tertiary alcohols react differently in certain reactions.

Practice 31.2

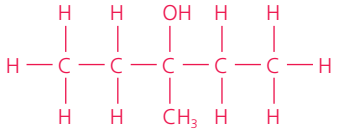
Draw the structure for each of the following alcohols and state whether it is primary, secondary or tertiary.

a) 3-methylpentan-3-ol

b) Butan-1-ol

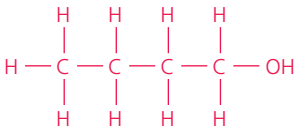
c) Cyclohexanol

a)



tertiary alcohol

b)



primary alcohol

c)



secondary alcohol

Substitution reactions of alcohols with halides

Alcohols react with hydrogen halides (HI, HBr and HCl) to form haloalkanes.

$$R-OH + HX \longrightarrow R-X + H_2O$$

The order of reactivity of alcohols in substitution reactions with halides is 3° > 2° > 1°.

The order of reactivity of the hydrogen halides is HI > HBr > HCl (HF does not react).

Reaction with hydrogen chloride

Prepare a chloroalkane by bubbling dry hydrogen chloride gas through the anhydrous alcohol in the presence of anhydrous zinc chloride as a catalyst.

For example,

$$CH_3CH_2CH_2OH(l) + HCl(g) \xrightarrow{ZnCl_2\ \text{catalyst}} \underset{\text{1-chloropropane}}{CH_3CH_2CH_2Cl(l)} + H_2O(l)$$

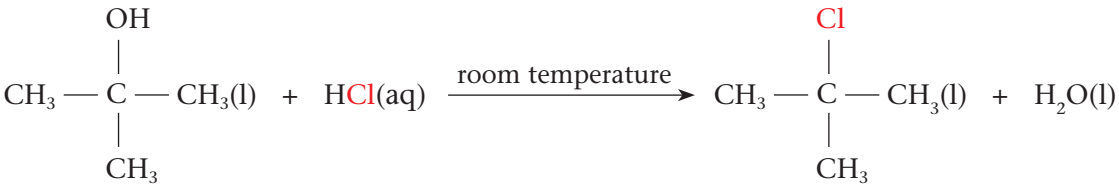
We can also prepare a chloroalkane by refluxing an alcohol with concentrated hydrochloric acid using zinc chloride as a catalyst.

For example,

$$CH_3CH_2CH_2OH(l) + HCl(aq) \xrightarrow[ZnCl_2 \text{ catalyst}]{\text{reflux}} CH_3CH_2CH_2Cl(l) + H_2O(l)$$

1-chloropropane

Tertiary alcohols react reasonably rapidly with concentrated hydrochloric acid without a catalyst. For example, methylpropan-2-ol reacts with concentrated hydrochloric acid at room temperature. 2-chloro-2-methylpropane is formed in the reaction.



2-chloro-2-methylpropane

Reaction with hydrogen bromide

Normally, the reaction is carried out by refluxing together a mixture of alcohol, sodium bromide and concentrated sulphuric acid.

Concentrated sulphuric acid reacts with sodium bromide to produce hydrogen bromide which reacts with the alcohol.

For example,

$$H_2SO_4(l) + NaBr(s) \longrightarrow NaHSO_4(s) + HBr(g)$$

$$CH_3CH_2OH(l) + HBr(g) \xrightarrow{\text{reflux}} CH_3CH_2Br(l) + H_2O(l)$$

bromoethane
(boiling point 38 °C)

The mixture is then warmed to distil off the bromoalkane (Fig. 31.4).

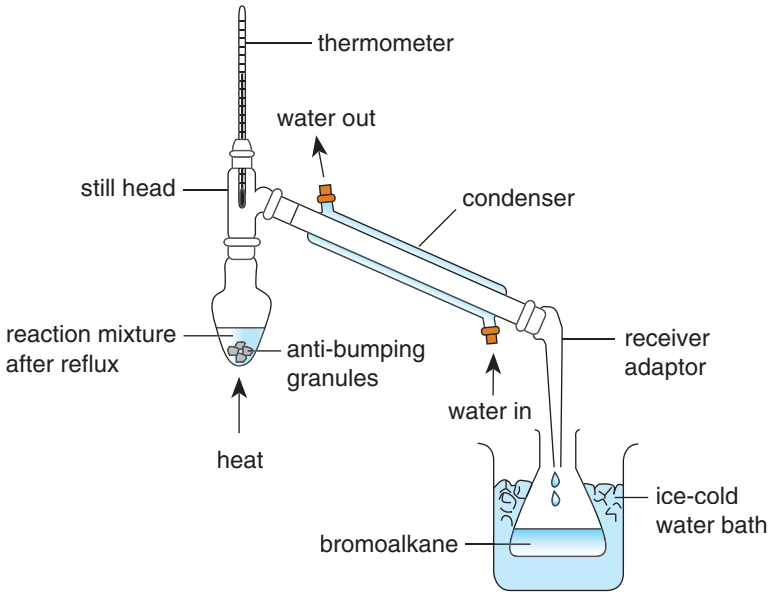


Fig. 31.4 A laboratory set-up for separating bromoalkane from the reaction mixture N6

N6 Examination questions often ask students to draw the experimental set-up for simple distillation.

Reaction with hydrogen iodide

In this case, the alcohol is reacted with a mixture of sodium iodide and concentrated phosphoric acid. Concentrated phosphoric acid reacts with sodium iodide to produce hydrogen iodide which reacts with the alcohol. The iodoalkane is distilled off.

Phosphoric acid is used instead of concentrated sulphuric acid because sulphuric acid oxidizes iodide ions to iodine and produces very little hydrogen iodide.

For example,

$$H_3PO_4(l) + NaI(s) \longrightarrow NaH_2PO_4(s) + HI(g)$$

$$CH_3CH_2OH(l) + HI(g) \xrightarrow{\text{reflux}} CH_3CH_2I(l) + H_2O(l)$$

iodoethane
(boiling point 72 °C)

Reaction with phosphorus halides

We can also convert alcohols to haloalkanes by reaction with phosphorus halides.

To prepare a bromoalkane or an iodoalkane, the alcohol is usually heated under reflux with a mixture of red phosphorus and bromine or iodine. Phosphorus first reacts with bromine or iodine to give phosphorus trihalide. The halide then reacts with the alcohol to give the corresponding haloalkane, which can be distilled off.

For example,

$$2P(s) + 3Br_2(l) \longrightarrow 2PBr_3(l)$$

$$3CH_3CH_2OH(l) + PBr_3(l) \xrightarrow{\text{reflux}} \underset{\text{bromoethane}}{3CH_3CH_2Br(l)} + \underset{\text{phosphorous acid}}{H_3PO_3(l)}$$

Phosphorus pentachloride reacts with alcohols in the cold.

The formation of hydrogen chloride fumes upon reaction with phosphorus pentachloride is a test for the presence of a hydroxyl group in a compound.

For example,

$$CH_3CH_2CH_2OH(l) + PCl_5(s) \xrightarrow{\text{room temperature}} \underset{\text{1-chloropropane}}{CH_3CH_2CH_2Cl(l)} + \underset{\text{phosphorus trichloride oxide}}{POCl_3(l)} + HCl(g)$$

Reaction with sulphur dichloride oxide (thionyl chloride)

Sulphur dichloride oxide (thionyl chloride) has the molecular formula $SOCl_2$. It reacts with primary and secondary alcohols to produce chloroalkanes. Sulphur dioxide and hydrogen chloride are given off.

For example,

$$CH_3CH_2CH_2OH(l) + SOCl_2(l) \xrightarrow{\text{reflux}} CH_3CH_2CH_2Cl(l) + SO_2(g) + HCl(g)$$

Table 31.3 summarizes the substitution reactions of alcohols with halides.

Table 31.3

Summary of the substitution reactions of alcohols with halides

Reaction with	Reagent	Condition	Product
Chloride	conc. HCl + $ZnCl_2$ catalyst	reflux	chloroalkane
	PCl_5	room temperature	
	$SOCl_2$	reflux	
Bromide	NaBr + conc. H_2SO_4	reflux	bromoalkane
	red P + Br_2	reflux	
Iodide	NaI + conc. H_3PO_4	reflux	iodoalkane
	red P + I_2	reflux	

Practice 31.3

Predict the products of the following reactions:

a) reflux butan-2-ol with sodium iodide and concentrated phosphoric acid;
2-iodobutane

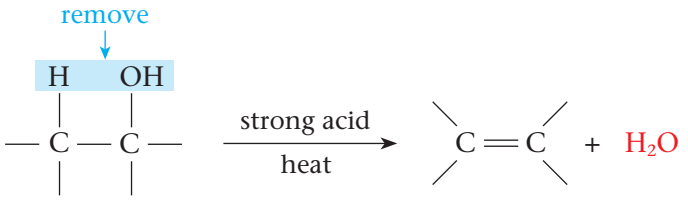
b) reflux cyclohexanol with red phosphorus and bromine;
bromocyclohexane

c) reflux pentan-1-ol with concentrated hydrochloric acid and zinc chloride catalyst.
1-chloropentane

Video
Preparing ethene and investigating its properties

Elimination reactions — dehydration of alcohols to form alkenes

Most alcohols dehydrate and form alkenes when they are heated in the presence of concentrated sulphuric acid or concentrated phosphoric acid.



AL 2009 Paper 2 Q4(b)(ii)

These reactions are described as **dehydration** since they involve the removal of a water molecule from a molecule of the reactant. Dehydration is an example of **elimination reaction**.

dehydration 脱水作用　　elimination reaction 消去反應

In an elimination reaction, atoms or groups of atoms are removed from two adjacent atoms (usually carbon atoms) of the reactant molecule.

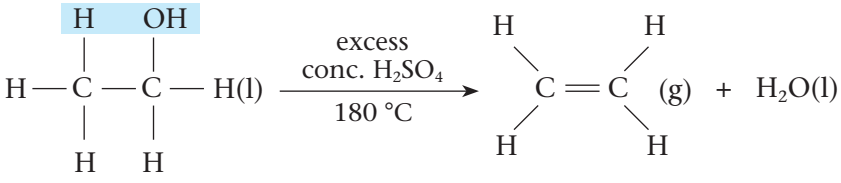
The experimental conditions — temperature and acid concentration — that are required to bring about dehydration are closely related to the structure of individual alcohols.

The relative ease with which alcohols undergo dehydration shows the following order:

Ease of dehydration: 3° > 2° > 1° alcohol

Dehydration of primary alcohols

Primary alcohols are the most difficult to dehydrate. For example, we can carry out the dehydration of ethanol by heating ethanol at 180 °C with excess concentrated sulphuric acid. Ethene is evolved and can be collected over water (Fig. 31.5).



ethanol (1° alcohol) ethene

We can regard the concentrated sulphuric acid as a dehydrating agent, removing water from the ethanol.

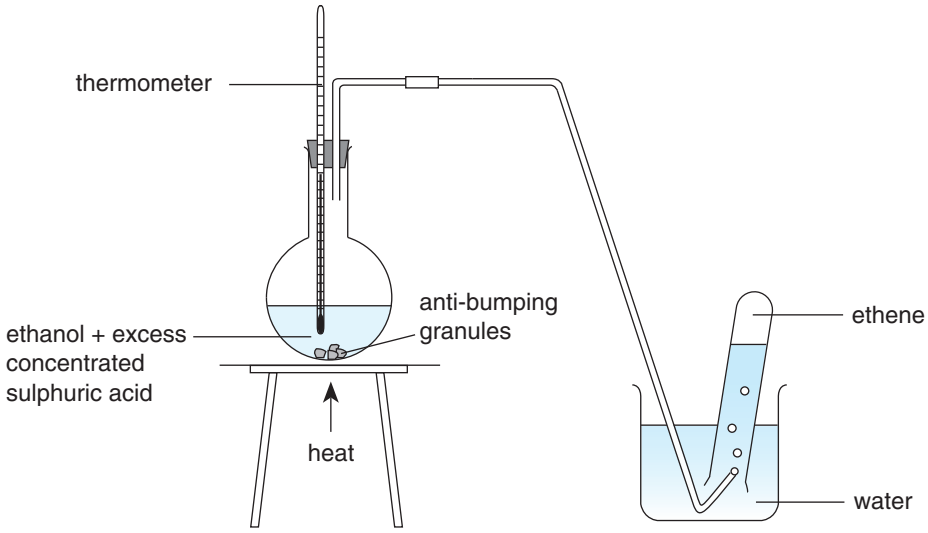


Fig. 31.5 Experimental set-up for the dehydration of ethanol using concentrated sulphuric acid

Alternatively, the hot ethanol vapour can be passed over heated aluminium oxide (Fig. 31.6).

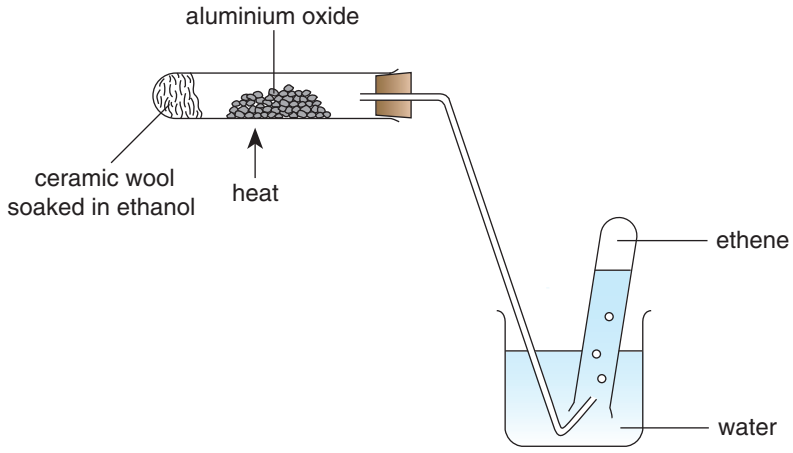
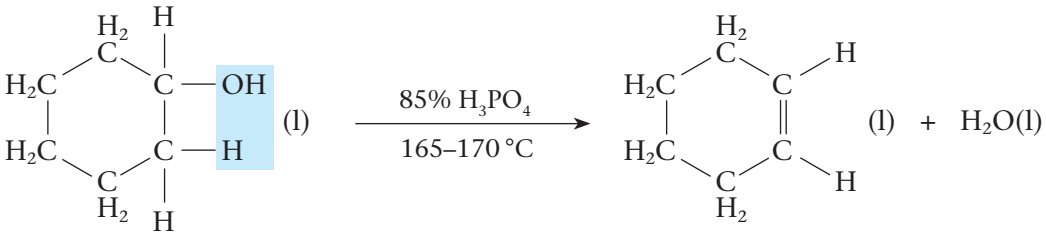


Fig. 31.6 Experimental set-up for the dehydration of ethanol using hot aluminium oxide

Dehydration of secondary alcohols

Secondary alcohols usually dehydrate under milder conditions. Cyclohexanol, for example, dehydrates in 85% phosphoric acid at 165–170 °C.

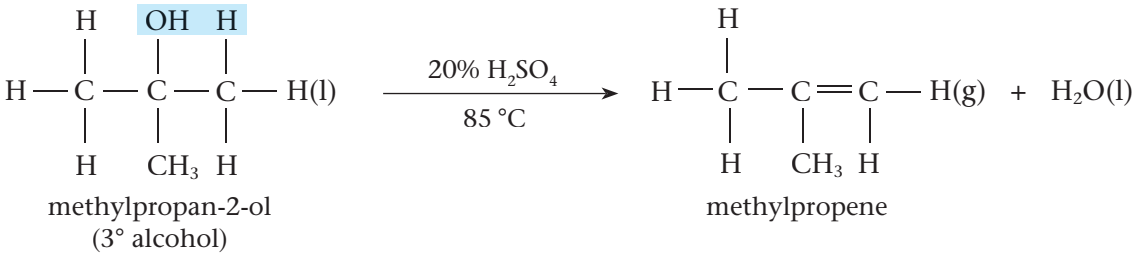


cyclohexanol
(2° alcohol)

cyclohexene

Dehydration of tertiary alcohols

Tertiary alcohols are usually so easily dehydrated that extremely mild conditions can be used. For example, methylpropan-2-ol dehydrates in 20% sulphuric acid at 85 °C.

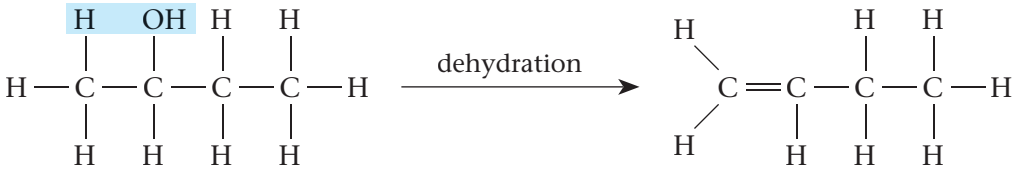


methylpropan-2-ol (3° alcohol) → methylpropene

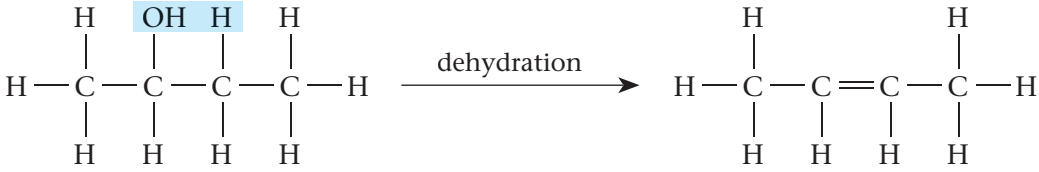
Major dehydration product of butan-2-ol

Dehydration of an alcohol may produce more than one alkene. Normally the most-substituted alkene is the major product.

For example, the dehydration of butan-2-ol produces but-1-ene and but-2-ene.



but-1-ene (minor product)



but-2-ene (major product)

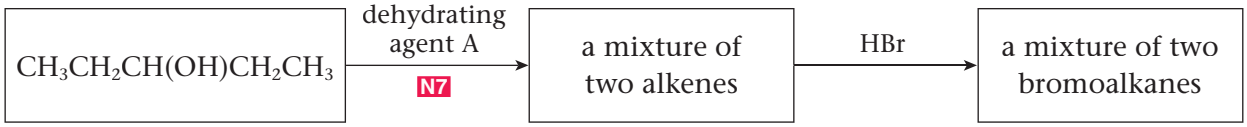
Remember that but-2-ene has *cis* and *trans* isomers. We have discussed this in Unit 30.

But-2-ene is the major dehydration product. But-1-ene has only one alkyl group attached to the carbon atoms of the C=C bond while but-2-ene has two. But-2-ene is the more substituted alkene and hence the major product of the dehydration.

Example 31.1

N7 Examination questions often ask about the structures of products when an alcohol is dehydrated.

Q Consider the conversion of the carbon compounds shown below:



X Y Z

a) Suggest what dehydrating agent A might be.

b) The mixture Y contains two alkenes with the same structural formula.

i) Write the structural formulae of these two alkenes.

ii) State the isomeric relationship of these two alkenes.

DSE 2013 Paper 1B Q15(c)

c) The mixture Z contains two bromoalkanes, one of which is optically active.

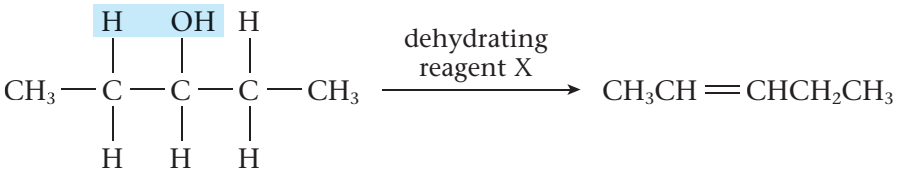
i) Draw the structures of these two bromoalkanes.

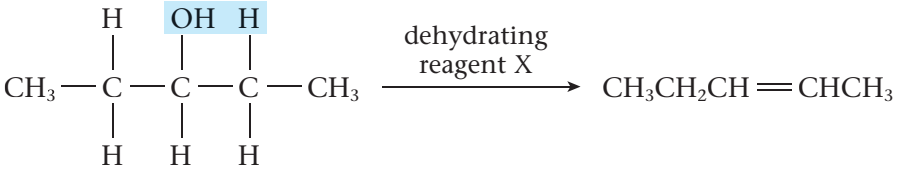
ii) Identify the bromoalkane which is optically active. Explain your choice.

DSE 2013 Paper 1B Q15(d)

A a) Concentrated sulphuric acid or phosphoric acid.

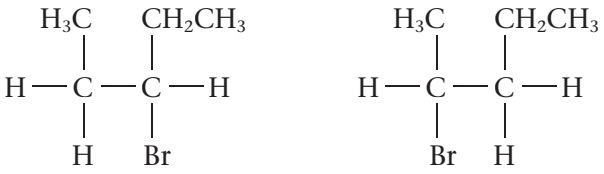
b) i)



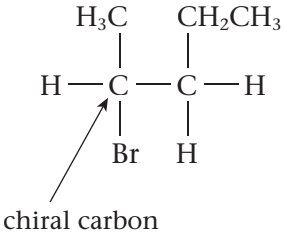


ii) The two alkenes are *cis* and *trans* isomers.

c) i)



ii)



is chiral and thus optically active.

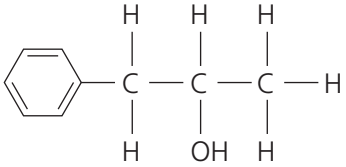
Practice 31.4

1 a) Draw the structures of alkenes that can be made by dehydration of $(CH_3)_2C(OH)CH_2CH_3$.
Refer to page T36.

b) Which of the alkenes drawn in (a) is likely to be the major product? Explain your answer.
Refer to the next page.

2 Alcohol A has the following structure:

DSE 2012 Paper 1A Q33



When A was heated with concentrated sulphuric acid, a mixture of three alkenes X, Y and Z were formed. X, Y and Z are isomers of each other. Only a small amount of Z was formed.

a) Name the type of reaction involved.
Dehydration / elimination

b) Draw the structures of alkenes X, Y and Z.
Refer to the next page.

c) Which of the alkenes drawn in (b) is likely to be the minor product Z? Explain your answer.
Refer to the next page.

d) What is the isomeric relationship between the other two alkenes, X and Y? *Cis-trans* isomers

Oxidation of alcohols

N8 Examination questions often ask about the oxidation of alcohols:
- the oxidizing agent required;
- the name(s) of the product(s).

31.1
Studying the properties of alcohols.

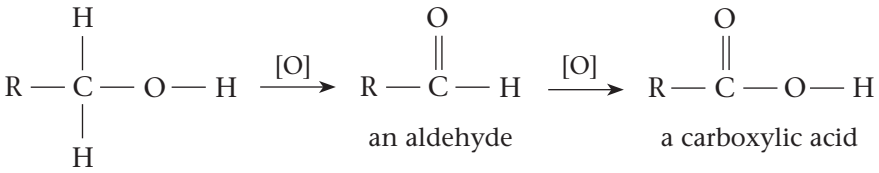
We have learnt that alcohols can be classified as primary, secondary or tertiary. During the oxidation of alcohols, primary, secondary and tertiary alcohols react differently.

A number of oxidizing agents can be used in the oxidation process, such as acidified potassium dichromate solution, acidified or alkaline potassium permanganate solution. N8

Oxidation of a carbon compound usually corresponds to increasing its oxygen content or decreasing its hydrogen content.

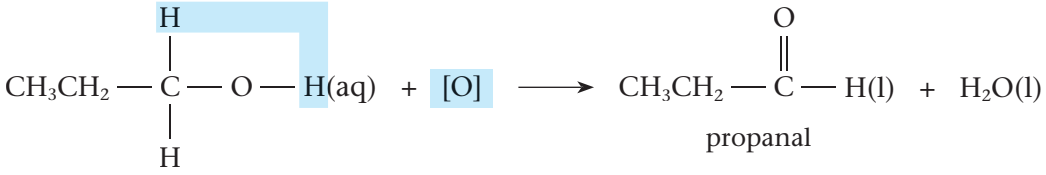
Oxidation of primary alcohols to aldehydes and carboxylic acids

Primary alcohols are easily oxidized to form aldehydes, and these aldehydes are rapidly oxidized to carboxylic acids.



a primary alcohol — an aldehyde — a carboxylic acid

Consider the oxidation of propan-1-ol by acidified potassium dichromate solution. The removal of two hydrogen atoms produces a new functional group — the aldehyde group, CHO.



propan-1-ol propanal

where [O] represents the oxygen atom from an oxidizing agent

Look at the ionic half-equation for the oxidation of a primary alcohol. The alcohol loses two electrons in the process.

$RCH_2OH(aq) \rightarrow RCHO(l) + 2H^+(aq) + 2e^-$

The electrons supplied by the alcohol reduce the dichromate ion.

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

The ionic equation for the oxidation process is:

$3RCH_2OH(aq) + Cr_2O_7^{2-}(aq) + 8H^+(aq) \rightarrow 3RCHO(l) + 2Cr^{3+}(aq) + 7H_2O(l)$

The propanal formed is rapidly oxidized to propanoic acid. To obtain propanal, we must choose the reaction conditions to prevent any subsequent oxidation of propanal to propanoic acid. We can either

- carry out the reaction at room temperature and add the oxidizing agent to the alcohol so that the oxidizing agent is never in excess; or
- heat the reaction mixture to distil off the propanal once formed so that it will not be further oxidized.

Table 31.4 shows the boiling points of propan-1-ol and the products of oxidation.

Table 31.4

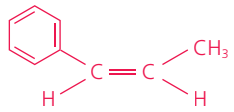
The boiling points of propan-1-ol and the products of oxidation

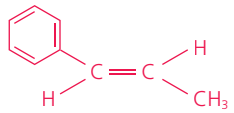
Compound	Boiling point (°C)
Propan-1-ol	97
Propanal	49
Propanoic acid	141

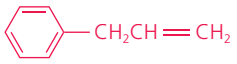
Practice 31.4

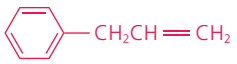
1 b) Alkene X has two alkyl groups attached to the carbon atoms of the carbon-carbon double bond while alkene Y has three. Alkene Y is the more substituted alkene and hence the major product.

2 b)







c)  is the minor product Z.
This alkene has only one substituent (benzyl group) attached to the carbon atoms of the carbon-carbon double bond while the other two are the more substituted alkenes, with phenyl group and methyl group attached to the carbon atoms of the carbon-carbon double bond, and hence the major dehydration products.

Using the experimental set-up shown in Fig. 31.7, the propan-1-ol is oxidized to propanal. Dichromate ions are reduced to chromium(III) ions in the process. Thus, the colour of the reaction mixture changes from orange to green (Fig. 31.8).

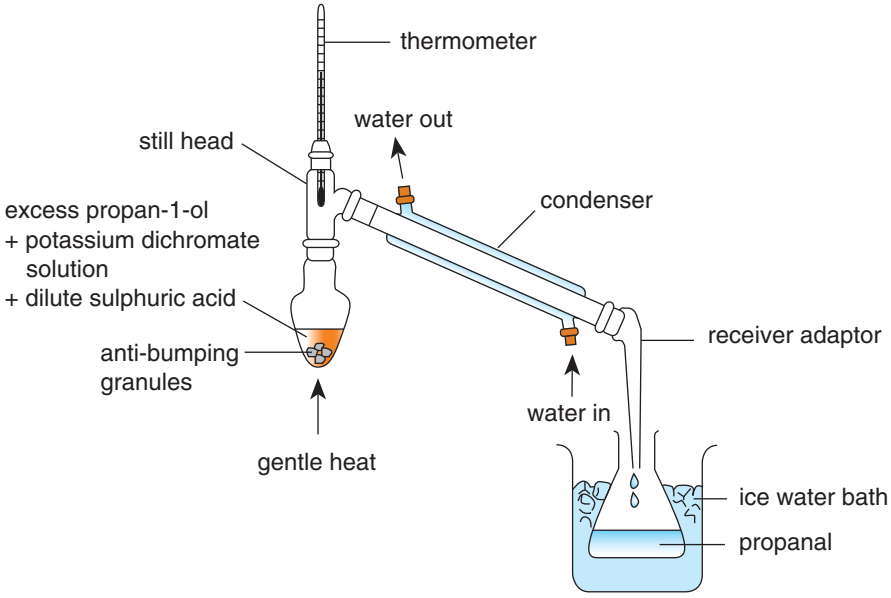


Fig. 31.7 An experimental set-up for the oxidation of propan-1-ol to propanal



Fig. 31.8 Left: Acidified potassium dichromate solution
Right: After the oxidation of propan-1-ol, the solution becomes green

Providing that the temperature on the thermometer is above 49 °C and below about 90 °C, the distillate is propanal. It is removed from the reaction mixture before it has the chance to be oxidized further.

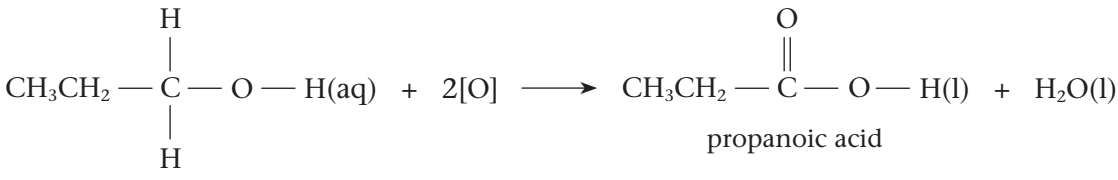
Oxidation of primary alcohols to carboxylic acids

31.2
Oxidizing ethanol to ethanoic acid and testing the ethanoic acid produced.

Video
Oxidizing ethanol

By changing the conditions of oxidation, it is possible for a primary alcohol molecule to lose two hydrogen atoms and gain one oxygen atom, producing a carboxylic acid molecule.

We can oxidize propan-1-ol to propanoic acid by refluxing it with acidified potassium dichromate solution (Fig. 31.9a).



CE 2009 Paper 1 Q8(a)

Reference Website
Refer to the following website for principles and applications of an alcohol breathalyzer:
http://www.howstuffworks.com/gadgets/automotive/breathalyzer3.htm
(accessed July 2014)

In the reflux condenser, all the vapour is condensed back into the flask. This prevents any loss of the reaction mixture and favours the oxidation of propan-1-ol to propanoic acid rather than to propanal.

To separate propanoic acid from the reaction mixture, transfer the mixture to the apparatus as shown in Fig. 31.9b. On heating, propanoic acid and water distil out to give an aqueous solution of propanoic acid.

CE 2009 Paper 1 Q8(c)(ii)

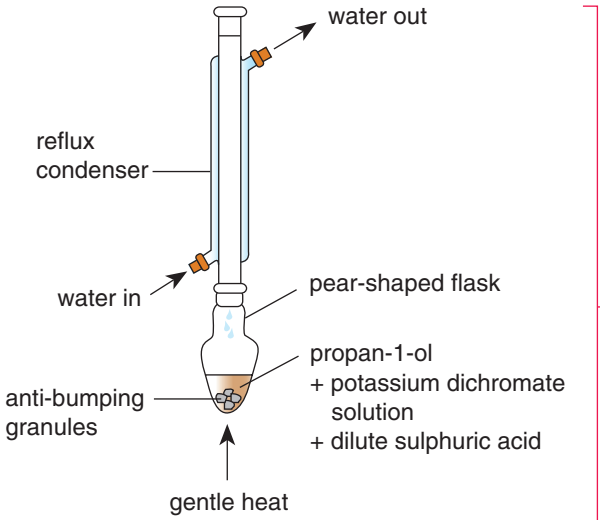


Fig. 31.9a An experimental set-up for the oxidation of propan-1-ol to propanoic acid №

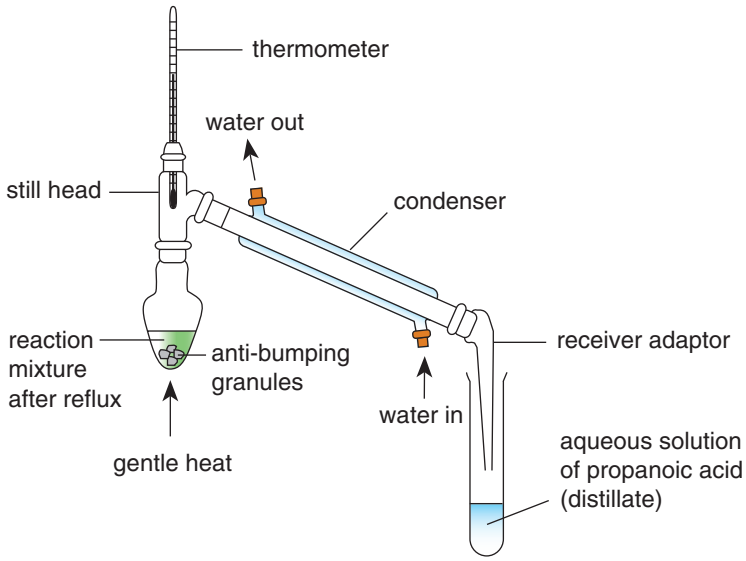


Fig. 31.9b An experimental set-up for separating propanoic acid from the reaction mixture

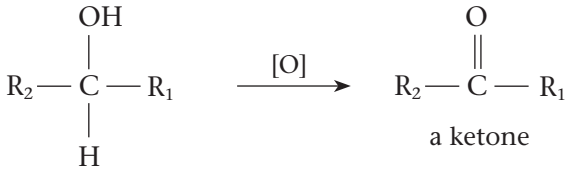
CE 2009 Paper 1 Q8(c)(i)
CE 2010 Paper 2 Q7
ASL 2009 Paper 1 Q7(c)
AL 2009 Paper 1 Q8(c)

№ Examination questions may ask students to draw the experimental set-up for heating under reflux. Remind students NOT to confuse this with the set-up for fractional distillation.

Acidified potassium permanganate solution is an oxidizing agent which is too powerful to stop at the aldehyde; it oxidizes primary alcohols to acids and oxidizes secondary alcohols to ketones.

Oxidation of secondary alcohols to ketones

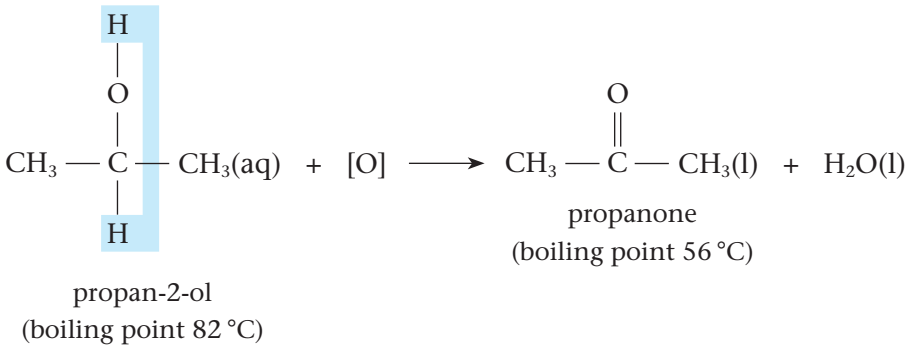
Secondary alcohols can be oxidized to give ketones.



a secondary alcohol

a ketone

For example, we can oxidize propan-2-ol to propanone by refluxing it with acidified potassium dichromate solution.

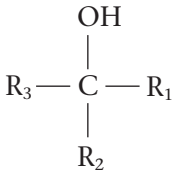


The reaction involves the removal of two hydrogen atoms from a propan-2-ol molecule.

Oxidation of tertiary alcohols

Tertiary alcohols are not oxidized by common oxidizing agents such as acidified potassium dichromate solution.

Look at the general structure of a tertiary alcohol:



There is no hydrogen atom attached to the carbon atom bearing the –OH group. Hence the oxidizing agent [O] cannot remove hydrogen atoms so that water and a carbonyl group are formed.

Distinguishing primary, secondary and tertiary alcohols

Since the three classes of alcohols give different reactions with acidified potassium dichromate solution, they can be distinguished from one another by the colour changes (Fig. 31.10) and products in such reactions (Table 31.5).

Table 31.5

Colour change and products of reactions between alcohols and acidified potassium dichromate solution

Alcohol	Warming with acidified potassium dichromate solution and distilling off the product		Heating under reflux with acidified potassium dichromate solution	
	Colour change of reaction mixture	Product	Colour change of reaction mixture	Product
Primary	orange to green	aldehyde	orange to green	carboxylic acid
Secondary	orange to green	ketone	orange to green	ketone
Tertiary	stays orange	no reaction	stays orange	no reaction

CE 2006 Paper 1 Q2(d)
ASL 2008 Paper 1 Q5(d)

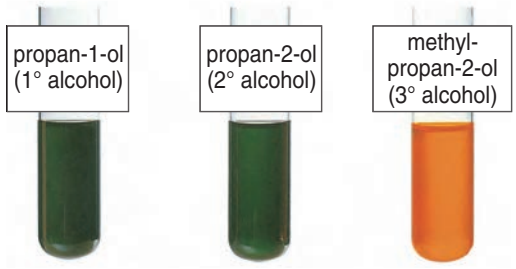


Fig. 31.10 During the oxidation of primary and secondary alcohols using acidified potassium dichromate solution, the colour of the reaction mixture changes from orange to green, but there is no colour change for tertiary alcohols

We will discuss this reaction later in this unit.

Fig. 31.11 summarizes the common reactions of propan-1-ol.

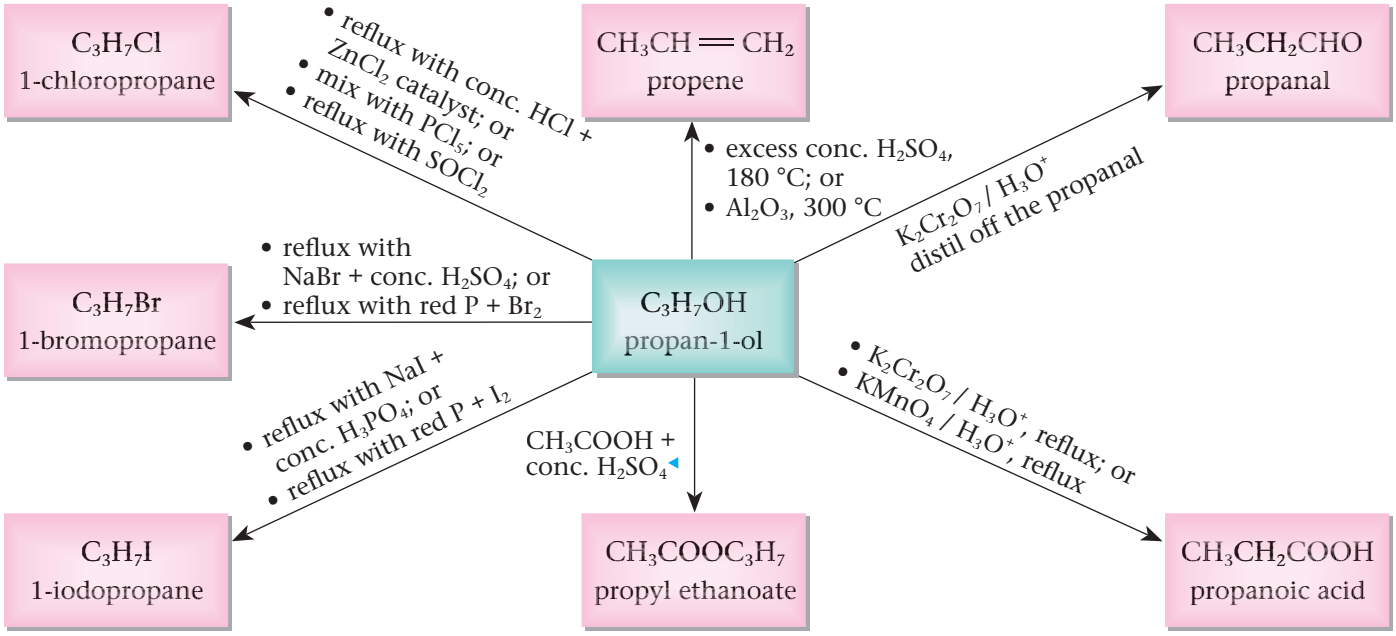
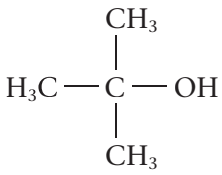


Fig. 31.11 A summary of the common reactions of propan-1-ol

Example 31.2

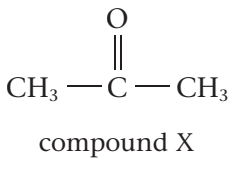
Q Suggest a chemical test to distinguish one compound from the other in each of the following pairs. Your answer should include the reagents used and the observations expected.

a)



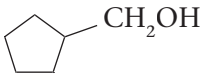
compound W

and



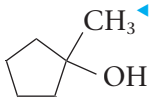
compound X

b)



compound Y

and



compound Z

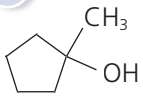
ASL 2006 Q9(a)(ii)

A a) Mix each compound with phosphorus pentachloride.

Only compound W gives steamy fumes.

b) Warm each compound with acidified potassium dichromate solution.

Only compound Y (a primary alcohol) turns the orange dichromate solution green.


is a tertiary alcohol as there are three carbons attached to the carbon bearing the –OH group.

Example 31.3

Q Compound X contains carbon, hydrogen and oxygen only. The relative molecular mass of X is 88.0. Complete combustion of 1.32 g of X gives 3.30 g of carbon dioxide and 1.62 g of water.

a) Deduce the molecular formula of X.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

b) Two chemical tests are performed on X and the results are as follows:

Test 1: X gives steamy fumes when mixed with phosphorus pentachloride.

Test 2: X gives a negative result when warmed with acidified potassium dichromate solution.

i) With reference to the result of *Test 1*, suggest a functional group that X has.

ii) Draw a possible structure of X. Explain your answer.

Continued on next page

A a) Mass of carbon in 1.32 g of X

$= 3.30 \text{ g} \times \frac{12.0}{44.0}$

$= 0.900 \text{ g}$

Mass of hydrogen in 1.32 g of X

$= 1.62 \text{ g} \times \frac{2.00}{18.0}$

$= 0.180 \text{ g}$

Mass of oxygen in 1.32 g of X
$= (1.32 - 0.900 - 0.180) \text{ g}$
$= 0.240 \text{ g}$

	Carbon	Hydrogen	Oxygen
Mass of element in the compound	0.900 g	0.180 g	0.240 g
Number of moles of atoms that combine	$\frac{0.900 \text{ g}}{12.0 \text{ g mol}^{-1}} = 0.0750 \text{ mol}$	$\frac{0.180 \text{ g}}{1.0 \text{ g mol}^{-1}} = 0.180 \text{ mol}$	$\frac{0.240 \text{ g}}{16.0 \text{ g mol}^{-1}} = 0.0150 \text{ mol}$
Simplest ratio of atoms	$\frac{0.0750 \text{ mol}}{0.0150 \text{ mol}} = 5.00$	$\frac{0.180 \text{ mol}}{0.0150 \text{ mol}} = 12.0$	$\frac{0.0150 \text{ mol}}{0.0150 \text{ mol}} = 1.00$

$\therefore$ the empirical formula of X is $C_5H_{12}O$.

Let $(C_5H_{12}O)_n$ be the molecular formula of X.

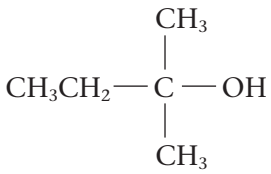
Relative molecular mass of X
$= n(5 \times 12.0 + 12 \times 1.0 + 16.0)$
$= 88n$

i.e. $88n = 88.0$
$n = 1$

$\therefore$ the molecular formula of X is $C_5H_{12}O$.

b) i) X contains a hydroxyl group (–OH).

ii) A possible structure of X:



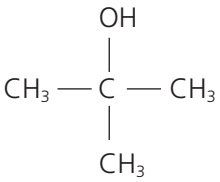
X is a tertiary alcohol because it resists the oxidation by acidified potassium dichromate solution.

Practice 31.5

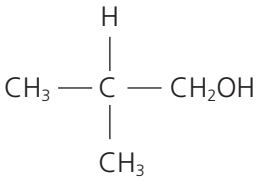
Four isomers with the molecular formula C_4H_9OH are shown below.

$CH_3CH_2CH_2CH_2OH$

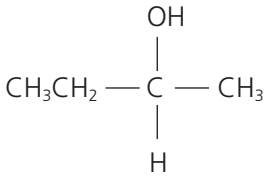
isomer A



isomer B



isomer C



isomer D

a) Identify the isomer which is resistant to oxidation by acidified potassium dichromate solution.

Isomer B

b) i) Identify the isomer which can be oxidized to a ketone. Give the structure of the ketone formed.

ii) In an experiment, the isomer is oxidized to ketone by using acidified potassium dichromate solution.

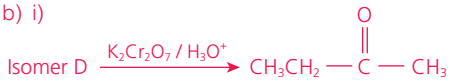
What would you observe as the reaction proceeds?

The reaction mixture turns form orange to green.

c) Identify ONE of the isomers which can be oxidized to an aldehyde. Give the structure of the aldehyde formed.

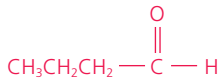
Practice 31.5

b) i)

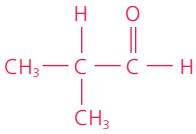


c) Any one of the following:

- Isomer A



- Isomer C



Problem Solving

Add acidified potassium dichromate solution to each liquid and warm.

Only CH_3CH_2OH and $CH_3(CH_2)_7OH$ change the colour of the dichromate solution from orange to green.

Mix CH_3CH_2OH and $CH_3(CH_2)_7OH$ with water separately. CH_3CH_2OH is miscible with water so a homogenous layer is formed. $CH_3(CH_2)_7OH$ is immiscible with water so two layers are formed.

Mix the other two liquids with phosphorus pentachloride separately.

Only $(CH_3)_3COH$ gives steamy fumes.

Worksheet

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Problem Solving

A, B, C and D are four unlabelled bottles, each containing one of the following colourless liquids:

CH_3CH_2OH $(CH_3)_3COH$ $CH_3(CH_2)_2Br$ $CH_3(CH_2)_7OH$

Suggest how you can identify the contents of each bottle.

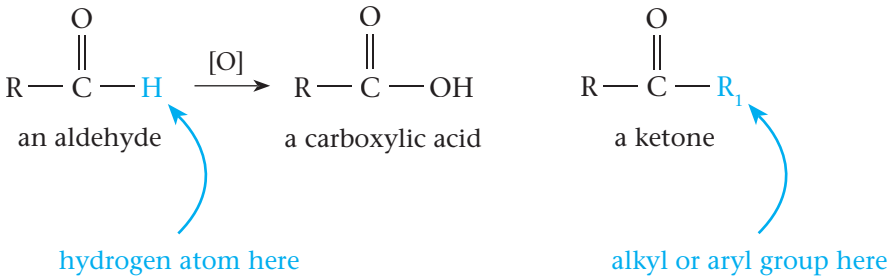
31.6 Reactions of aldehydes and ketones

Many reactions of aldehydes and ketones are similar because they both contain the carbonyl group  . However, they differ in their reactions with oxidizing and reducing agents.

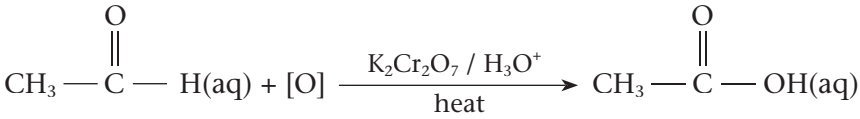
31.3
Studying the properties of propanal and propanone.

Oxidation of aldehydes and ketones

Aldehydes are easily oxidized to yield carboxylic acids, but ketones are generally inert towards oxidation. This difference is a consequence of structure: aldehydes contain a hydrogen atom next to the carbonyl group. This hydrogen atom is readily oxidized to –OH. Hence aldehydes are readily oxidized.



Aldehydes can be oxidized by acidified potassium dichromate solution. For example, when we warm acidified potassium dichromate solution with ethanal, ethanal is oxidized to ethanoic acid. The solution changes from orange to green as the dichromate ions are reduced to chromium(III) ions.



On the other hand, ketones are not oxidized by acidified potassium dichromate solution.

Reduction of aldehydes and ketones

We have learnt that aldehydes and ketones can be produced by the oxidation of alcohols. Partial oxidation of a primary alcohol produces an aldehyde, and oxidation of a secondary alcohol produces a ketone.

We use [H] to represent a reducing agent.

These reactions can be reversed by reduction — aldehydes are readily reduced to primary alcohols, and ketones to secondary alcohols.



In the laboratory, aldehydes and ketones can be reduced by either

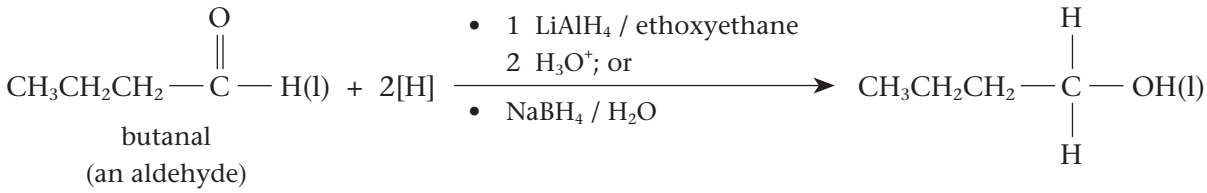
- lithium tetrahydridoaluminate (lithium aluminium hydride) ($LiAlH_4$); or
- sodium tetrahydridoborate (sodium borohydride) ($NaBH_4$).

Lithium tetrahydridoaluminate reacts violently with water, and therefore reductions with this reagent must be carried out in anhydrous conditions, usually in anhydrous ethoxyethane. The final product is hydrolyzed by a dilute acid.

By contrast, sodium tetrahydridoborate reductions can be carried out in water. Sodium tetrahydridoborate is a less powerful reducing agent than lithium tetrahydridoaluminate. It is usually chosen because of its safety and ease of handling.

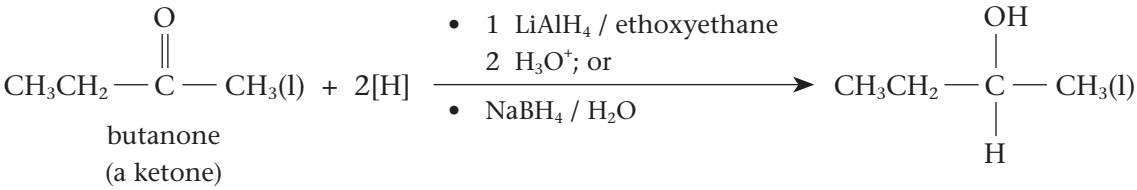
For example,

AL 2012 Paper 2 Q13(b)(ii)



butanal
(an aldehyde)

butan-1-ol
(a primary alcohol)



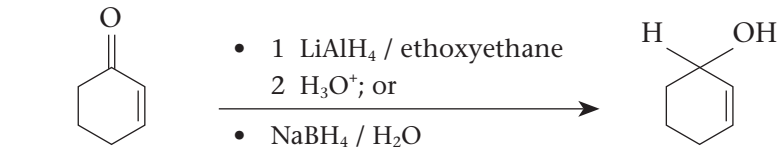
butanone
(a ketone)

butan-2-ol
(a secondary alcohol)

DSE 2013 Paper 1B Q15(b)

The above two hydride reagents do not affect carbon-carbon double bonds. Hence they can convert unsaturated aldehydes and ketones into unsaturated alcohols.

For example,



cyclohex-2-enone
(a ketone)

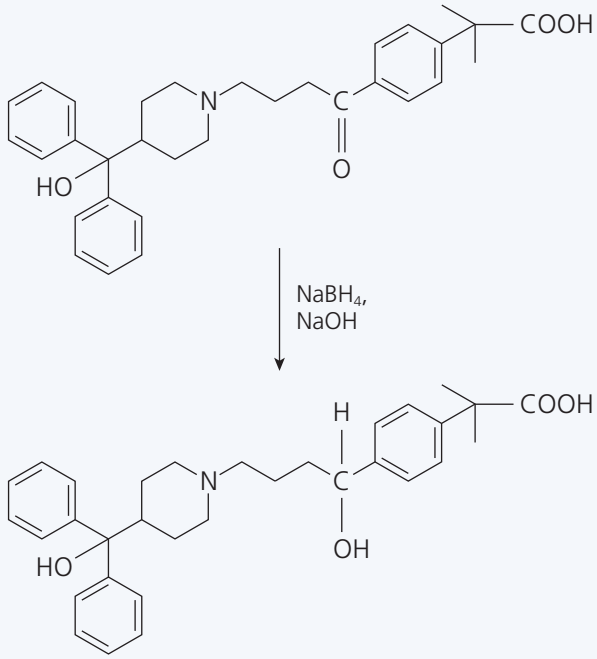
cyclohex-2-enol
(a secondary alcohol)

Do you know

Manufacture of fexofenadine

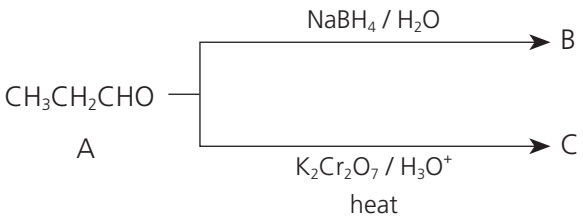
Fexofenadine hydrochloride is an antihistamine drug used in the treatment of hay fever and other allergy conditions. The manufacturing process involves the reduction reaction shown on the right.

Why is sodium tetrahydridoborate used rather than the more powerful reducing agent lithium tetrahydridoaluminate? This is because the latter also reduces carboxylic acids. In this reaction it would reduce the $-COOH$ group to $-CH_2OH$ group.



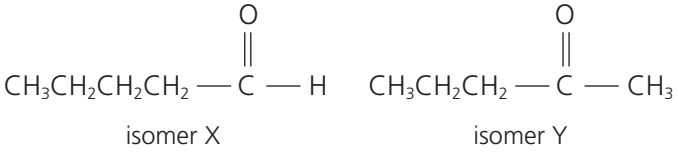
Practice 31.6

1 Compound A can be converted into two different carbon compounds as shown below. Give the structural formulae of the new compounds B and C.



B: $CH_3CH_2CH_2OH$
C: CH_3CH_2COOH

2 The structures of two structural isomers with the molecular formula $C_5H_{12}O$ are shown below.



a) Name the type of structural isomerism involved.

Functional group isomerism

b) Suggest a chemical test that can be used to distinguish the two isomers.

State what you would observe with each isomer.

Warm each isomer with acidified potassium dichromate solution.

Isomer X turns the orange dichromate solution green.

There is no observable change for isomer Y.

N10 Refer to page T36.

31.4
Studying the properties of ethanoic acid.

Esterification reactions proceed very slowly in the absence of strong acids.

31.7 Reactions of carboxylic acids N10

Reaction with alkalis and hydrogencarbonates

Reaction with alkalis

Like any acid, a carboxylic acid reacts with an alkali to form a salt and water.

DSE 2012 Paper 1A Q34

For example,

$$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COO^- Na^+(aq) + H_2O(l)$$

Reaction with hydrogencarbonates

A carboxylic acid produces carbon dioxide gas when added to a sodium hydrogencarbonate solution.

CE 2008 Paper 2 Q47
CE 2010 Paper 2 Q38

For example,

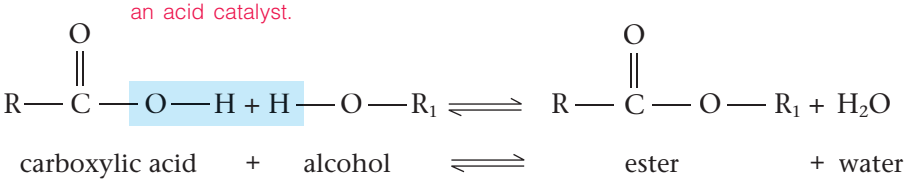
$$CH_3COOH(aq) + NaHCO_3(aq) \longrightarrow CH_3COO^- Na^+(aq) + H_2O(l) + CO_2(g)$$

Esterification

CE 2010 Paper 1 Q3
DSE 2012 Paper 1A Q34
ASL 2009 Paper 1 Q7(a)
AL 2009 Paper 1 Q8(a)

A carboxylic acid reacts with an alcohol in the presence of concentrated sulphuric acid (as a catalyst) in a reversible reaction to form an ester. N11

N11 Esters can also be produced by the reaction of acid chlorides (also called acyl chlorides) with alcohols. The reaction occurs rapidly and does not require an acid catalyst.



The reaction is called **esterification**. It can be considered as a condensation reaction. N12 N13

✓ A **condensation reaction** is a reaction in which two or more molecules react together to form a larger molecule with the elimination of a small molecule such as water.

Do you know

Fragrance of esters

Natural esters are responsible for many flower scents and fruit flavours. The artificial fruit flavours (e.g. banana, peach) used in some confectionery products are made by mixing synthetic esters together.

The table below shows the fragrance of some synthetic esters and the reactants needed for making the esters.

Fragrance of synthetic ester	Reactants needed to make the ester
Banana or pear	propan-1-ol and ethanoic acid
Pineapple	ethanol and butanoic acid
Peach	phenylmethanol and ethanoic acid

esterification 酯化作用　　condensation reaction 縮合反應

N12 Esterification cannot go to completion.

N13 Refer to page T38.

Reference Website

Refer to the following website for an animation showing an esterification reaction:
http://resources.hkedcity.net/resource_detail.php?rid=1478721266
(accessed July 2014)

Video

Reacting ethanol with ethanoic acid

The trivial name for ethyl ethanoate is ethyl acetate.

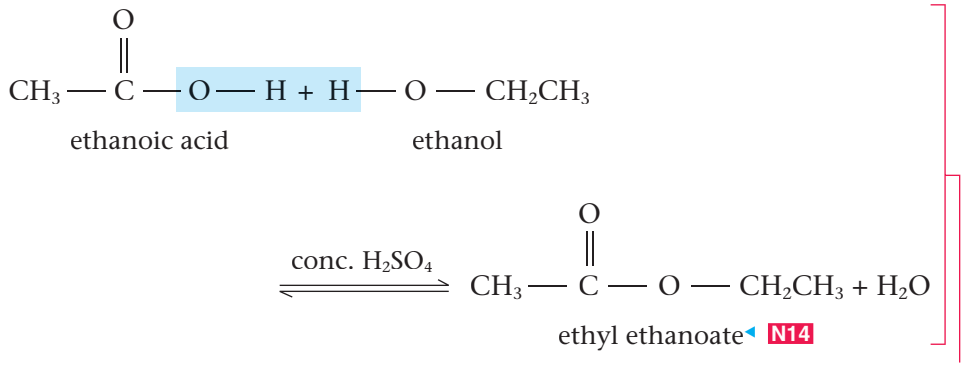
N14 Refer to page T38.

N15 The ester can also be separated from the reaction mixture by fractional distillation.

An ester is also formed by heating a carboxylic acid and an alcohol in the presence of concentrated sulphuric acid under reflux.

CE 2008 Paper 1 Q9

For example,



N14

CE 2006 Paper 1 Q2(a)
CE 2007 Paper 2 Q16
ASL 2005 Q3(b)(i)

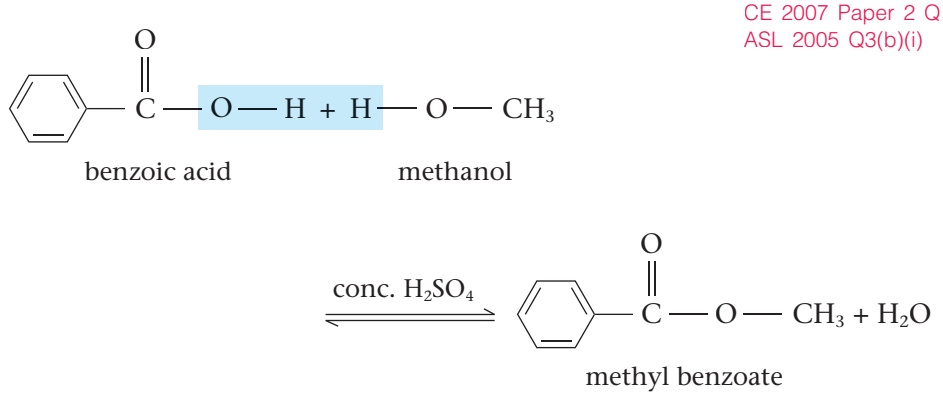


Fig. 31.12 shows an experiment in which ethanol and ethanoic acid are warmed in the presence of concentrated sulphuric acid. The reaction mixture is poured into sodium carbonate solution after heating. An insoluble layer of ethyl ethanoate forms on the solution. A sweetish smell can be detected. N15

CE 2006 Paper 1 Q2(e)
CE 2007 Paper 2 Q16
ASL 2009 Paper 1 Q7(a)
AL 2009 Paper 1 Q8(a)

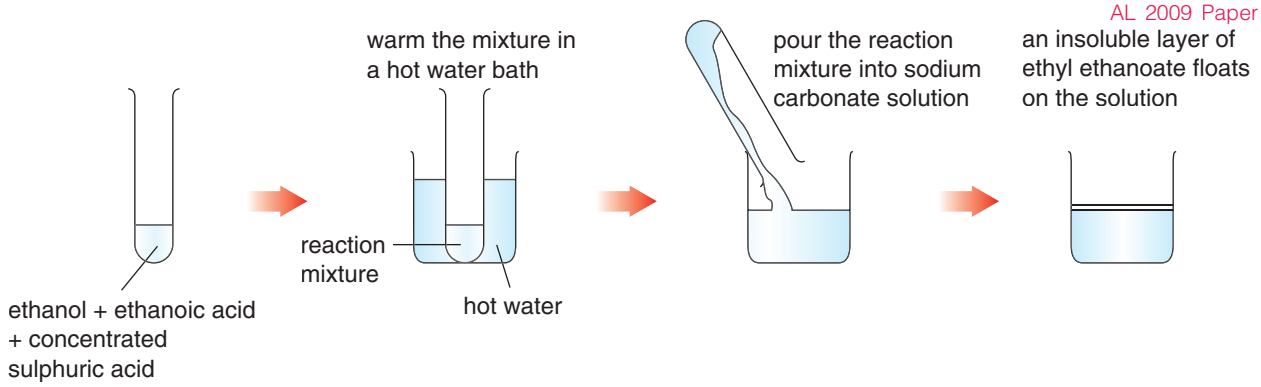


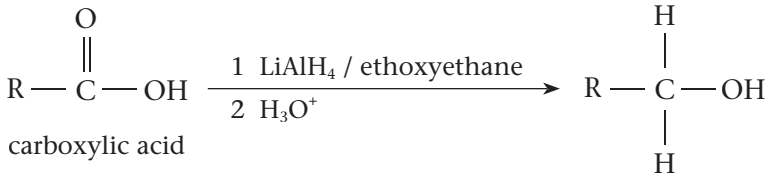
Fig. 31.12 Ethyl ethanoate is formed when ethanol and ethanoic acid undergo esterification

Sodium tetrahydridoborate ($NaBH_4$) is a less powerful reducing agent. It cannot reduce carboxylic acids.

Reduction

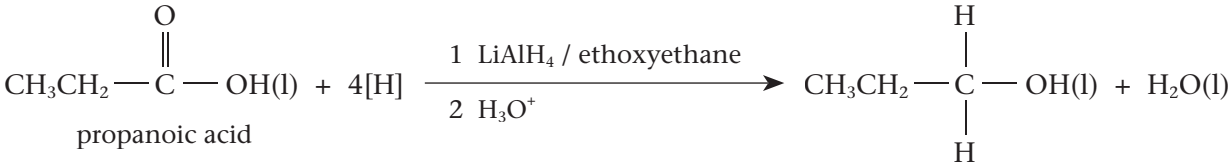
DSE 2012 Paper 1A Q34

A carboxylic acid can be reduced to a primary alcohol by the powerful reducing agent lithium tetrahydridoaluminate ($LiAlH_4$). The final product has to be hydrolyzed by a dilute acid.

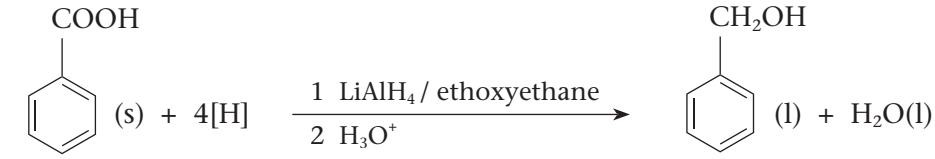


carboxylic acid — primary alcohol

For example,



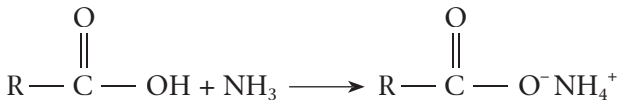
propanoic acid — propan-1-ol



benzoic acid — phenylmethanol

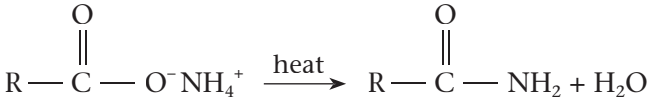
Amides from carboxylic acids

Carboxylic acids react with aqueous ammonia to form ammonium salts.



an ammonium carboxylate

If we evaporate the water and subsequently heat the dry salt, dehydration produces an amide.



an amide

We can also produce ethanamide from ethanoic acid by adding solid ammonium carbonate to an excess of concentrated ethanoic acid.

$$2CH_3COH(l) + (NH_4)_2CO_3(s) \longrightarrow 2CH_3CO^-NH_4^+(s) + CO_2(g) + H_2O(l)$$

When the reaction is complete, the mixture is heated under reflux. The ammonium salt dehydrates and produces ethanamide.

$$CH_3CO^-NH_4^+(s) \longrightarrow CH_3CNH_2(s) + H_2O(l)$$

ethanamide

The excess ethanoic acid is to prevent the dissociation of the ammonium salt before it dehydrates.

The mixture is distilled at about 170 °C to remove the excess ethanoic acid and water — leaving almost pure ethanamide in the flask.

Fig. 31.13 summarizes some common reactions of propanoic acid.

The ammonium salt tends to split into ammonia and the parent acid on heating, and they recombine on cooling.

$CH_3COONH_4(s) \rightleftharpoons CH_3COOH(l) + NH_3(g)$

The presence of the excess ethanoic acid helps to prevent this from happening by shifting the position of equilibrium to the left.

We will discuss chemical equilibrium in Topic 11 Chemical Equilibrium.

31.5

Identifying unknown carbon compounds.

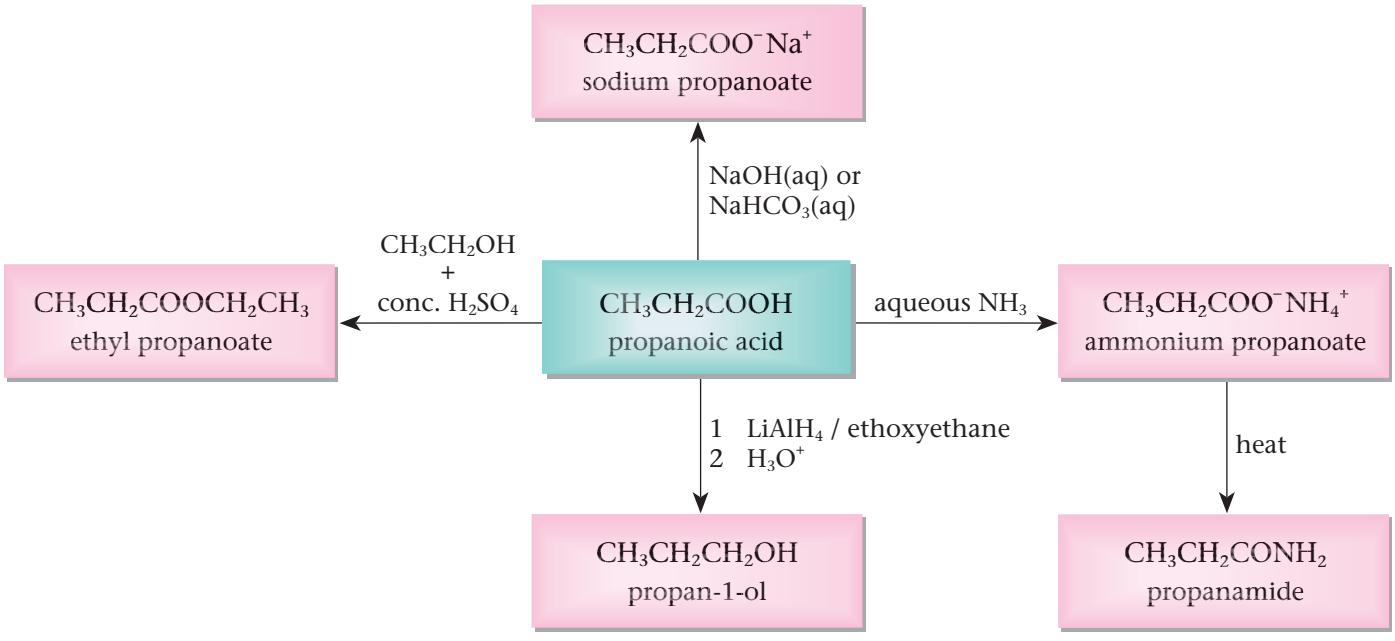
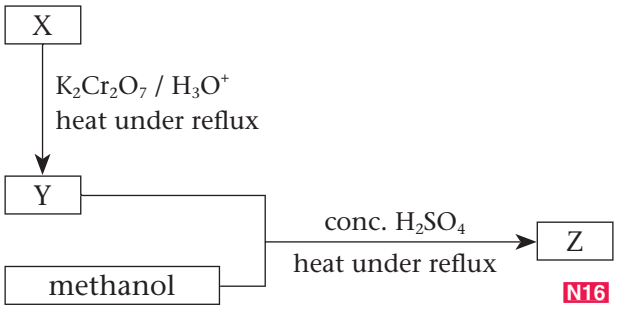


Fig. 31.13 Summary of some common reactions of propanoic acid

Example 31.4

Q X, Y and Z are carbon compounds. The flow diagram below shows the conversion of X to Z.



N16

CE 2006 Paper 1 Q2(a)

a) Z has a pleasant smell and its molecular formula is $C_4H_8O_2$.

i) Draw the structure of Z.

ii) Write a chemical equation for the reaction between Y and methanol.

iii) State the function of concentrated sulphuric acid in the reaction of Y with methanol.

CE 2006 Paper 1 Q2(d)

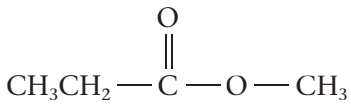
iv) Suggest ONE reason why Y and methanol are heated under reflux.

b) i) Give the systematic name of X.

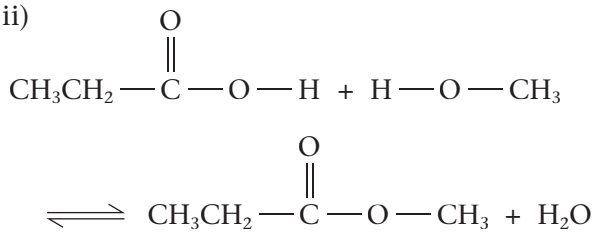
ii) State the expected observation when X reacts with acidified potassium dichromate solution.

CE 2006 Paper 1 Q2(c) & (d)

A a) i)



ii)



iii) As a catalyst

iv) Most carbon compounds have low boiling points. Heating under reflux can reduce the loss of reactants / products by evaporation.

b) i) Propan-1-ol / propanal

ii) The colour of the reaction mixture changes from orange to green.

N16 Examination questions often ask about the preparation of an ester.

- Students should be able to state clearly the conditions required for esterification to take place, heating a carboxylic acid with an alcohol in the presence of concentrated sulphuric acid.
- Concentrated sulphuric acid acts as a catalyst, NOT as an oxidizing agent.
- Given the names of a carboxylic acid and an alcohol, students should be able to deduce the systematic name / structure of the ester formed, or vice versa.
- Students should be able to draw the experimental set-up for heating carboxylic acid and alcohol under reflux.
- Remind students to distinguish between the preparation of an ester and the oxidation of an alcohol. The preparation of an ester requires a catalyst (concentrated sulphuric acid) while the oxidation of an alcohol requires an oxidizing agent (e.g. acidified potassium dichromate solution).

Example 31.5

Q Compound X has the molecular formula $C_4H_6O_2$.

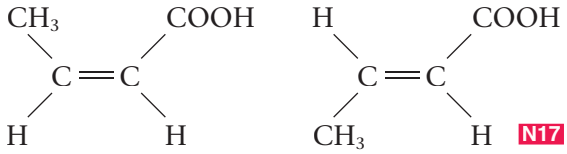
- Effervescence occurred when X was added to sodium carbonate solution.
- When aqueous bromine was shaken with X, the aqueous bromine turned colourless quickly.
- Compound X exhibits *cis-trans* isomerism.

Use the information above to draw the structures of the two possible isomers of compound X.

A

- X gave effervescence with sodium carbonate solution. This shows the presence of a –COOH group in X.
- X decolourized aqueous bromine quickly. This shows the presence of a carbon-carbon multiple bond in X.

The structures of the two possible isomers of X are as follows:



N17

N17 Refer to page T39.

Practice 31.7

1 The following paragraph was extracted from the laboratory report of a student on the preparation of a carbon compound X.

> A mixture of ethanoic acid, methanol and a few drops of concentrated sulphuric acid in a test tube was warmed in a water bath for a few minutes. The contents of the test tube was then poured into a beaker of sodium carbonate solution.

a) Why is a water bath, instead of a naked flame, used for heating the mixture in the test tube?
Methanol in the reaction mixture is flammable.

b) State TWO observable changes when the mixture of the test tube was poured into the sodium carbonate solution. An insoluble layer can be observed.
A fruity smell is detected.

c) i) Give the structural formula and systematic name of carbon compound X prepared.

ii) Name the homologous series to which carbon compound X belongs.

2 Carbon compound Z contains carbon, hydrogen and oxygen only. The relative molecular mass of Z is 102.0. Analysis of Z gives the following results:
Refer to page T40.

Test 1: 1.53 g of Z contains 0.900 g of carbon and 0.150 g of hydrogen.

Test 2: There is no observable change when potassium carbonate solution is added to Z.

Test 3: The yellow-brown colour of aqueous bromine remains unchanged when several drops of aqueous bromine are added to Z.

a) Deduce the molecular formula of Z. $C_5H_{10}O_2$

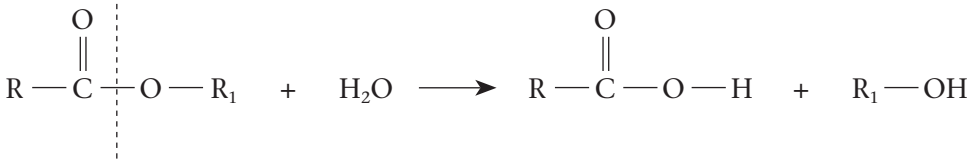
b) i) Suggest a possible structure of Z. Explain your answer.

ii) Give the IUPAC name of the compound represented by the structure you suggested in (i).

1 c) i) $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3$ methyl ethanoate
ii) Ester

31.8 Hydrolysis of esters

The esterification reaction between a carboxylic acid and an alcohol is reversible. The backward reaction, in which an ester splits up to reform the original acid and alcohol, is called hydrolysis — so called because water causes the compound to split up.

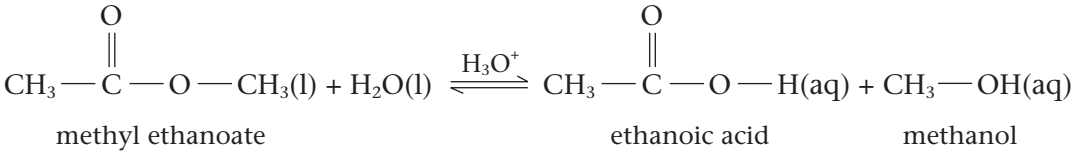


However, the reaction of the ester with water is very slow by itself. A catalyst is needed to speed up the reaction. The catalyst can be either a dilute acid (such as sulphuric acid) or a dilute sodium hydroxide solution.

Using an acid

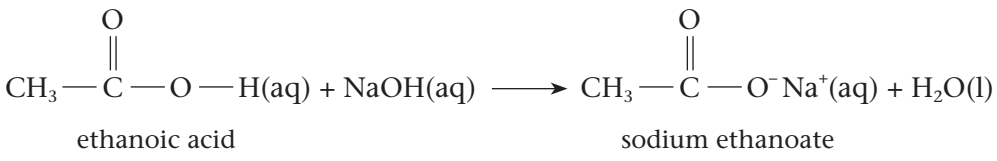
The ester is heated under reflux with dilute sulphuric acid.

For example, hydrolysis of methyl ethanoate produces ethanoic acid and methanol.



Using an alkali

Methyl ethanoate can also be hydrolyzed by boiling it with sodium hydroxide solution. This is a more efficient process because when the ethanoic acid is released, it is neutralized by the alkali to form a sodium salt. Salt formation removes the ethanoic acid from the reaction mixture. Thus, the hydrolysis goes to completion.



The overall reaction thus becomes:

$$\underset{\text{methyl ethanoate}}{CH_3COCH_3(l)} + NaOH(aq) \longrightarrow \underset{\text{sodium ethanoate}}{CH_3COO^-Na^+(aq)} + \underset{\text{methanol}}{CH_3OH(aq)}$$

In the laboratory, first heat a mixture of an ester and an aqueous alkali under reflux (Fig. 31.14a). Then separate the products by fractional distillation (Fig. 31.14b). The alcohol is obtained as distillate. The salt of the carboxylic acid remains in the solution as it is less volatile. We can obtain the free carboxylic acid by adding excess acid, such as hydrochloric or sulphuric acid, to the salt.

For example,

$$\underset{\substack{\text{ethanoate ion}\\\text{(from salt)}}}{CH_3COO^-(aq)} + \underset{\substack{\text{hydrogen ion}\\\text{(from acid)}}}{H^+(aq)} \rightleftharpoons \underset{\text{ethanoic acid}}{CH_3COOH(aq)}$$

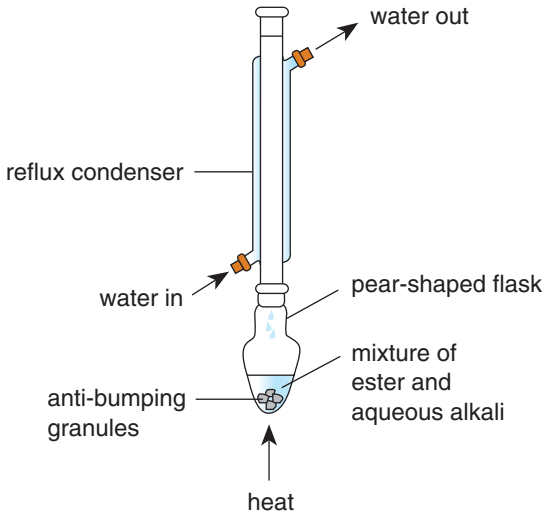


Fig. 31.14a An experimental set-up for the alkaline hydrolysis of an ester

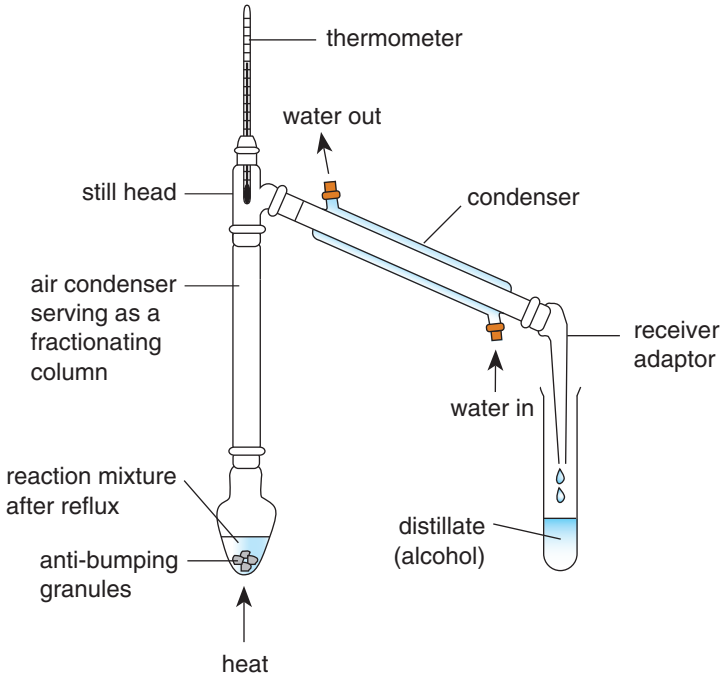


Fig. 31.14b Separating alcohol from the products formed in the hydrolysis of ester

Game
Typical reactions

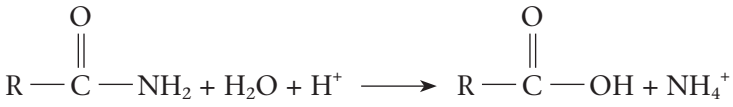
31.9 Hydrolysis of amides

Just as esters, the reaction of an amide with water is extremely slow, but the reaction can be catalyzed by an acid or an alkali. Heat the amide under reflux with an aqueous acid or alkali. Moderately concentrated solution of the acid or alkali is usually used.

DSE 2014 Paper 1B Q12(a)(i)
ASL 2009 Paper 1 Q8(a)(ii)

The products depend on whether an acid or alkali is used in the hydrolysis. If an acid is used, the product contains carboxylic acid and ammonium ions. If an alkali is used, the carboxylic acid loses H^+ ions and the product contains carboxylate ions and ammonia.

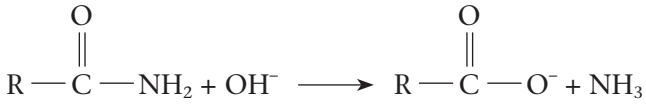
In acidic solution



For example, when boiling hydrochloric acid is used for hydrolysis, ethanamide forms ethanoic acid and ammonium chloride.

$$CH_3CONH_2(aq) + H_2O(l) + HCl(aq) \xrightarrow{\text{heat}} CH_3COOH(aq) + NH_4Cl(aq)$$

In alkaline solution



DSE 2013 Paper 1A Q32

When ethanamide is refluxed with sodium hydroxide solution, N18 ammonia gas is given off and we are left with a solution containing sodium ethanoate.

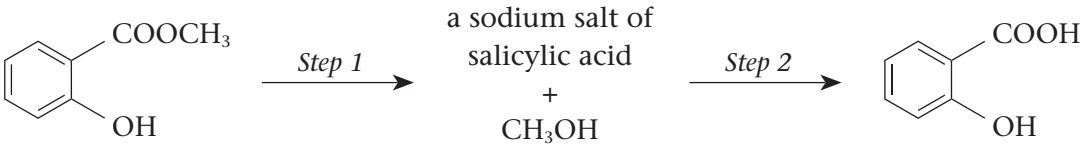
N18 In the hydrolysis, sodium hydroxide is a reactant, NOT a catalyst in the reaction.

$$CH_3CONH_2(aq) + NaOH(aq) \xrightarrow{\text{heat}} CH_3COO^-Na^+(aq) + NH_3(g)$$

We can obtain ethanoic acid by adding excess dilute acid to the solution containing sodium ethanoate.

Example 31.6

Q Oil of wintergreen is a common ester. Salicylic acid can be obtained from it in two steps.



oil of wintergreen — salicylic acid

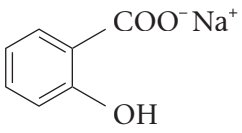
a) Give the reagent and condition used in *Step 1*.

b) Suggest a structure for the sodium salt of salicylic acid formed in *Step 1*.

c) Suggest a reagent that can be used in *Step 2*. DSE 2014 Paper 1B Q12(a)(iii)

A a) Sodium hydroxide solution

Heat under reflux

b) 

c) Dilute sulphuric acid / dilute hydrochloric acid

Practice 31.8

Propanoic acid can be obtained by heating ethyl propanoate with excess sodium hydroxide solution under reflux.

a) Write a chemical equation for the hydrolysis reaction.

b) Draw a labelled diagram of the set-up used for heating ethyl propanoate with excess sodium hydroxide solution under reflux.

c) Suggest how you can obtain propanoic acid from the reaction mixture after reflux.

a) $CH_3CH_2-\overset{\overset{O}{\|}}{C}-O-CH_2CH_3(l) + NaOH(aq) \longrightarrow CH_3CH_2-\overset{\overset{O}{\|}}{C}-O^-Na^+(aq) + CH_3CH_2OH(aq)$

b) Refer to page T41.

c) Separate the products by fractional distillation.
Ethanol is obtained as distillate. Sodium propanoate remains in the solution.
Add excess dilute sulphuric acid / hydrochloric acid to the sodium salt to obtain propanoic acid.

Have you mastered?

Key terms

combustion 燃燒 78 substitution reaction 取代反應 78 halogenation 鹵化作用 78

addition reaction 加成反應 79 hydrogenation 氫化作用 79 hydrolysis 水解作用 85

primary alcohol 一級醇 86 secondary alcohol 二級醇 86 tertiary alcohol 三級醇 86

sulphur dichloride oxide 氧二氯化硫 90 dehydration 脱水作用 91

elimination reaction 消去反應 91 lithium tetrahydridoaluminate 四氫合鋁酸鋰 106

sodium tetrahydridoborate 四氫合硼酸鈉 106 anhydrous ethoxyethane 無水乙氧基乙烷 106

esterification 酯化作用 108 condensation reaction 縮合反應 108

Checklist

After studying this unit, you should be able to

☐ describe the following reactions, in terms of reagents, reaction conditions and observations, and write the relevant chemical equations:

- alkanes — substitution with halogens;
- alkenes — addition reactions;
- haloalkanes — substitution with $OH^-(aq)$;
- alcohols — substitution with halides and oxidation;
- aldehydes and ketones — oxidation and reduction;
- carboxylic acids — reaction with alkalis and hydrogencarbonates, esterification, reduction and amide formation;
- esters — hydrolysis;
- amides — hydrolysis;

☐ predict and name the products of the above reactions.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Reactions of alkanes

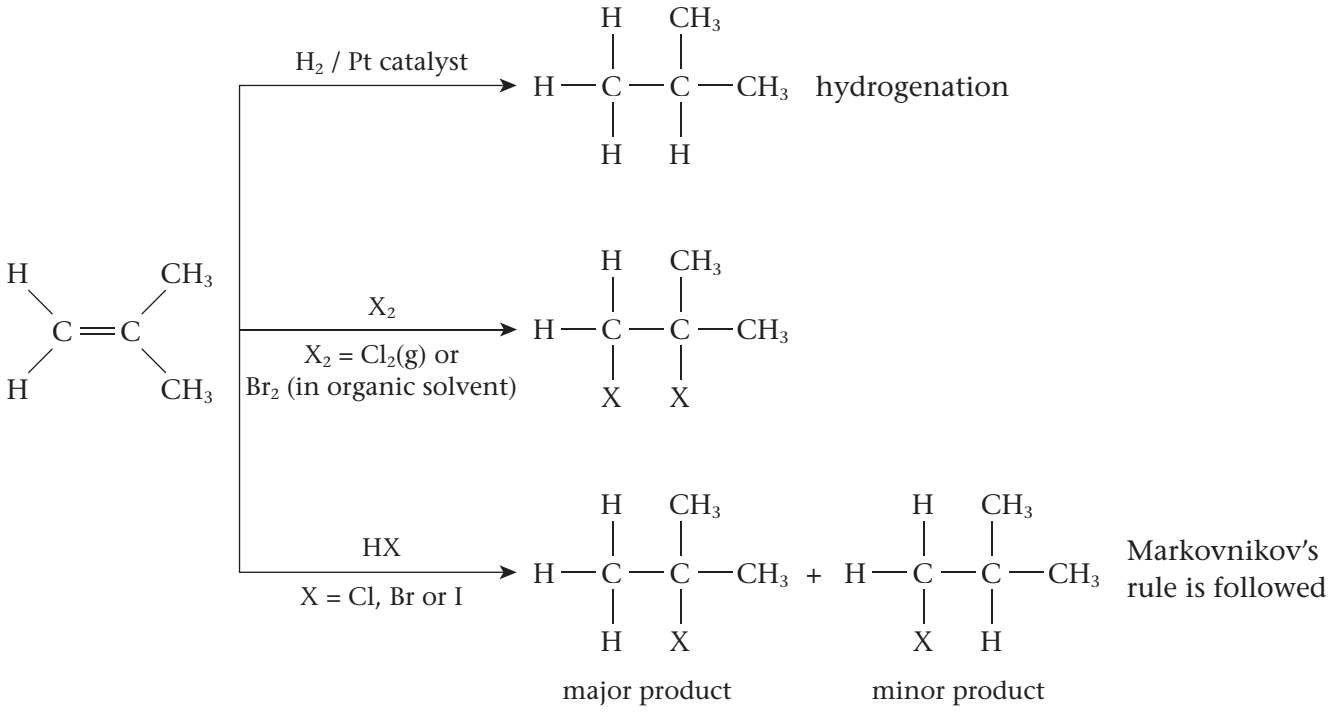
a) Combustion

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

b) Substitution reactions with halogens

$$CH_4 \xrightarrow[\text{UV light or heat}]{Cl_2} CH_3Cl + CH_2Cl_2 + CHCl_3 + CCl_4 + HCl$$

2 Addition reactions of alkenes



3 Markovnikov's rule for addition reaction of an asymmetric alkene:

When a molecule HA adds to an asymmetric alkene, the major product is the one in which the hydrogen atom attaches itself to the carbon atom already carrying the larger number of hydrogen atoms.

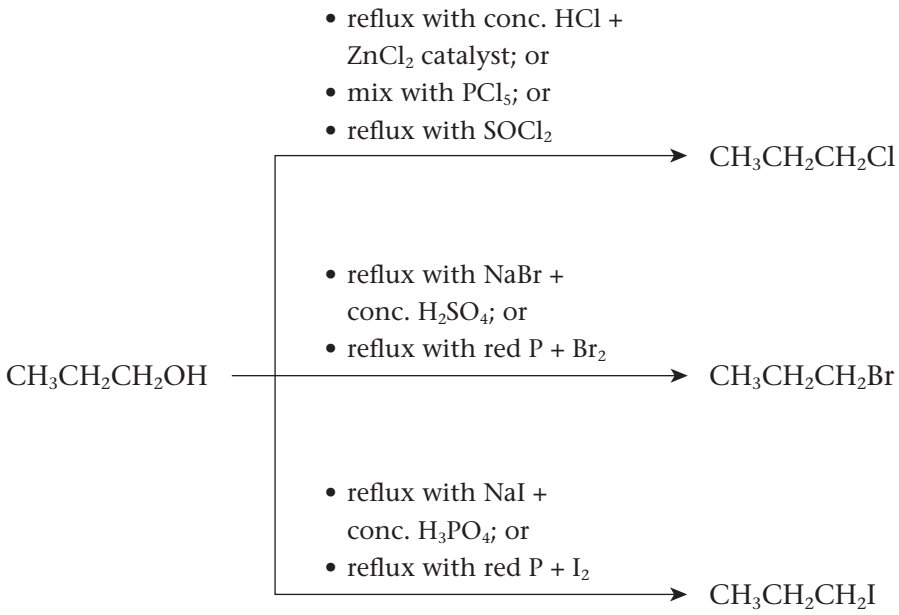
4 Substitution reactions of haloalkanes — alkaline hydrolysis of haloalkanes

$$R{-}X \xrightarrow[\text{reflux}]{NaOH(aq)} R{-}OH$$

Have you mastered?

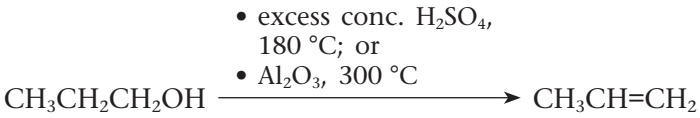
5 Substitution reactions of alcohols with halides

Order of reactivity of alcohols is 3° > 2° > 1°

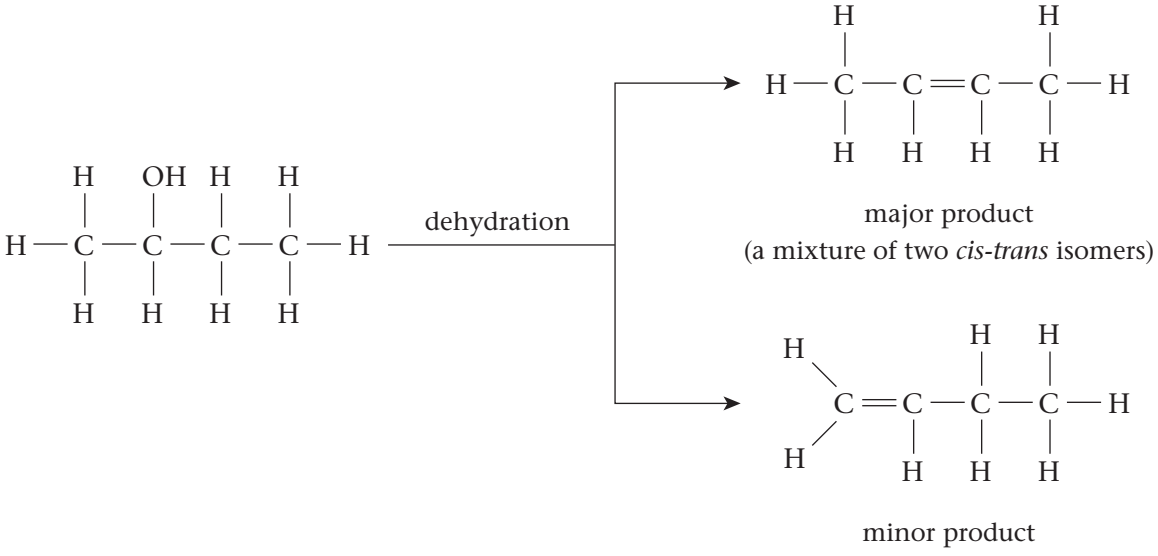


6 Dehydration of alcohols to form alkenes

a) Ease of dehydration: 3° > 2° > 1° alcohol



b) Dehydration of an alcohol may produce more than one alkene. Normally the most-substituted alkene is the major product.



Have you mastered?

7 Oxidation of alcohols

$$CH_3CH_2CH_2OH \xrightarrow[\text{gentle heat, distil off the product}]{K_2Cr_2O_7 \,/\, H_3O^+} CH_3CH_2CHO \xrightarrow[\text{further oxidation}]{\text{heat under reflux}} CH_3CH_2COOH$$

(1° alcohol)

$$CH_3CH(OH)CH_3 \xrightarrow[\text{heat under reflux}]{K_2Cr_2O_7 \,/\, H_3O^+} CH_3COCH_3$$

(2° alcohol)

$$(CH_3)_3COH \xrightarrow[\text{reflux}]{K_2Cr_2O_7 \,/\, H_3O^+} \text{no reaction}$$

(3° alcohol)

8 Oxidation and reduction of aldehydes and ketones

a) Oxidation of aldehydes

$$CH_3CH_2CHO \xrightarrow[\text{heat}]{K_2Cr_2O_7 \,/\, H_3O^+} CH_3CH_2COOH$$

b) Reduction of aldehydes and ketones

$$CH_3CH_2CHO \xrightarrow[\bullet\ NaBH_4 \,/\, H_2O]{\bullet\ 1\ LiAlH_4 \,/\, \text{ethoxyethane}\ \ 2\ H_3O^+;\ \text{or}} CH_3CH_2CH_2OH$$

aldehyde — 1° alcohol

$$CH_3COCH_3 \xrightarrow[\bullet\ NaBH_4 \,/\, H_2O]{\bullet\ 1\ LiAlH_4 \,/\, \text{ethoxyethane}\ \ 2\ H_3O^+;\ \text{or}} CH_3CH(OH)CH_3$$

ketone — 2° alcohol

9 Reactions of carboxylic acids

a) With alkalis

$$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COO^-Na^+(aq) + H_2O(l)$$

b) With hydrogencarbonates

$$CH_3COOH(aq) + NaHCO_3(aq) \longrightarrow CH_3COO^-Na^+(aq) + H_2O(l) + CO_2(g)$$

c) Esterification

$$CH_3-\overset{\overset{O}{\|}}{C}-OH + H-O-CH_2CH_3 \underset{}{\overset{\text{conc. } H_2SO_4}{\rightleftharpoons}} CH_3-\overset{\overset{O}{\|}}{C}-O-CH_2CH_3 + H_2O$$

Have you mastered?

d) Reduction

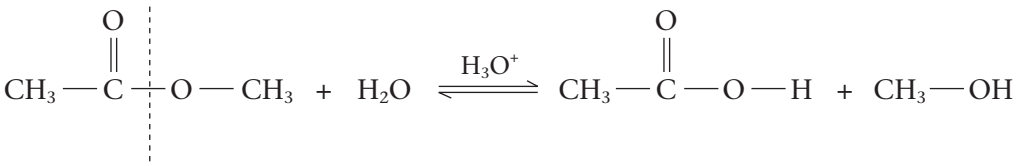
$$CH_3CH_2COOH \xrightarrow[2\ H_3O^+]{1\ LiAlH_4\ /\ \text{ethoxyethane}} CH_3CH_2CH_2OH$$

e) Amide synthesis

$$CH_3COOH + NH_3 \longrightarrow CH_3COO^-NH_4^+ \xrightarrow{\text{heat}} CH_3CONH_2 + H_2O$$

10 Hydrolysis of esters

a) Using an acid



b) Using an alkali



11 Hydrolysis of amides

a) Using an acid



b) Using an alkali



Have you mastered?

12 The following table summarizes the results of some tests for alcohols, aldehydes, ketones and carboxylic acids.

<table>
<tr><th>Test / Compound</th><th>Add water and test the pH of the solution</th><th>Warm with acidified potassium dichromate solution</th><th>Warm with ethanoic acid (with conc. sulphuric acid as a catalyst)</th><th>Warm with ethanol (with conc. sulphuric acid as a catalyst)</th><th>Mix with sodium hydrogen-carbonate solution</th></tr>
<tr><td>Primary alcohol</td><td rowspan="4">neutral</td><td>orange dichromate solution turns green</td><td rowspan="2">a sweet-smelling ester formed</td><td rowspan="3">no observable change</td><td rowspan="3">no observable change</td></tr>
<tr><td>Secondary alcohol</td><td rowspan="2">no observable change</td></tr>
<tr><td>Tertiary alcohol</td></tr>
<tr><td>Aldehyde</td><td>orange dichromate solution turns green</td><td rowspan="3">no observable change</td></tr>
<tr><td>Ketone</td><td rowspan="2">no observable change</td><td rowspan="2">a sweet-smelling ester formed</td><td rowspan="2">colourless CO_2 gas evolves</td></tr>
<tr><td>Carboxylic acid</td><td>pH about 3</td></tr>
</table>

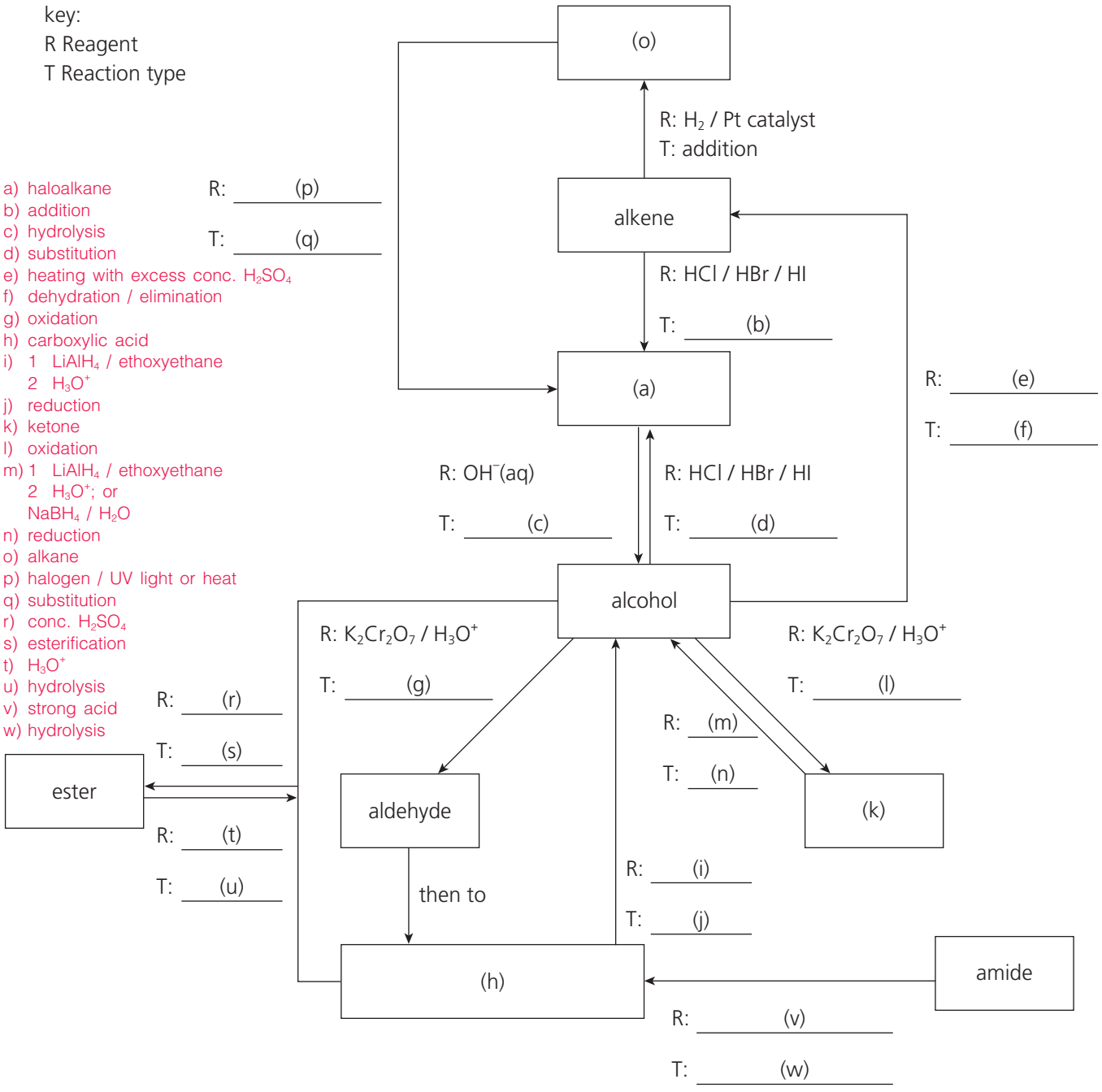
Have you mastered?

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the flow chart that includes reactions among the following carbon compounds: alkane, alkene, haloalkane, alcohol, aldehyde, ketone, carboxylic acid, ester and amide.



a) haloalkane
b) addition
c) hydrolysis
d) substitution
e) heating with excess conc. H_2SO_4
f) dehydration / elimination
g) oxidation
h) carboxylic acid
i) 1 $LiAlH_4$ / ethoxyethane
2 H_3O^+
j) reduction
k) ketone
l) oxidation
m) 1 $LiAlH_4$ / ethoxyethane
2 H_3O^+; or
$NaBH_4$ / H_2O
n) reduction
o) alkane
p) halogen / UV light or heat
q) substitution
r) conc. H_2SO_4
s) esterification
t) H_3O^+
u) hydrolysis
v) strong acid
w) hydrolysis

2 From the information given, draw the structure of each carbon compound. All of the compounds consist of molecules which have four carbon atoms.

a) A hydrocarbon that decolourizes aqueous bromine quickly.
$H_2C{=}CHCH_2CH_3$ or $CH_3CH{=}CHCH_3$ or $(CH_3)_2C{=}CH_2$

b) A compound that is oxidized to a ketone.
$CH_3CH(OH)CH_2CH_3$

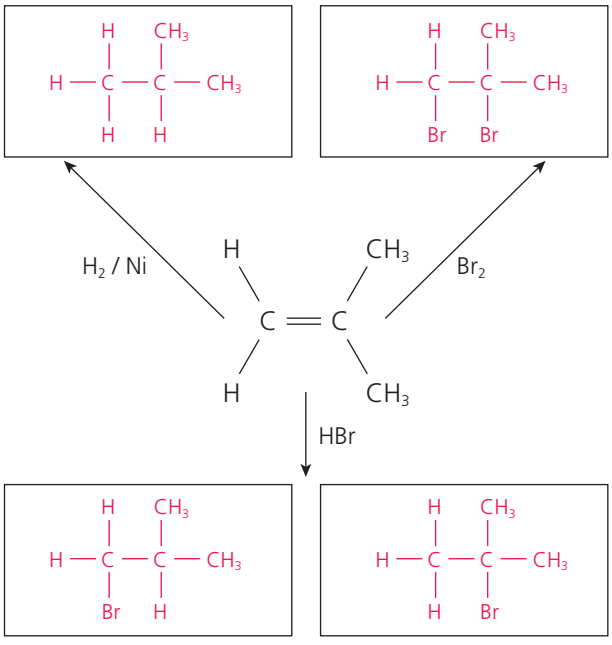
c) An ester that is formed from ethanoic acid.
$CH_3COOCH_2CH_3$

d) An aldehyde that gives butan-1-ol upon reduction.
$CH_3CH_2CH_2CHO$

e) An amide.
$CH_3CH_2CH_2CONH_2$ or $(CH_3)_2CHCONH_2$

3 Alkenes are unsaturated hydrocarbons used in the industrial production of many organic compounds.

Complete the flowchart below to show the organic product formed in each addition reaction of methylpropene.

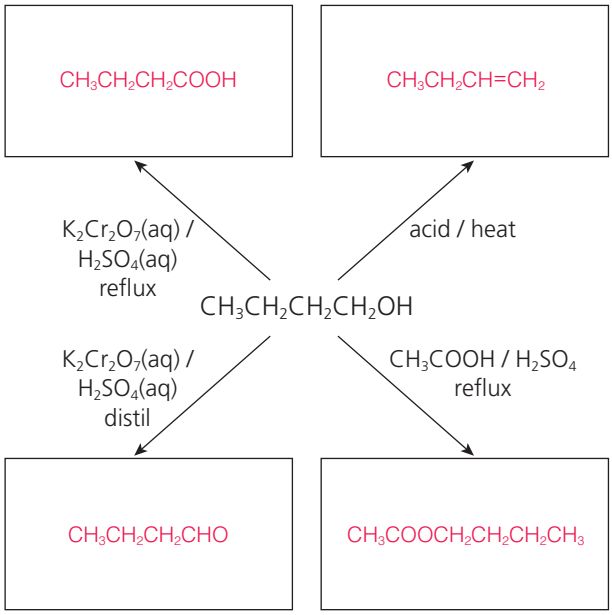


mixture of isomers

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2009, 3(a))

4 Alcohols are used in the industrial production of many organic compounds.

Complete the flowchart below to show the organic product formed in each of the reactions of butan-1-ol.



(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2010, 3(a))

5 Tiglic acid is a compound that is used as a defensive agent by some beetles.

Tiglic acid contains a carbon-carbon double bond and a carboxylic acid group.

Suggest ONE test for each of these groups in tiglic acid. State what you would do and what you would see as a positive result for the tests.

(Edexcel Advanced GCE, Unit 5, Feb. 2012, 22(a)(ii))

Test for carbon-carbon double bond
Any one of the following:
- Add aqueous bromine.
  The aqueous bromine changes from yellow-brown to colourless quickly.
- Add cold acidified dilute potassium permanganate solution.
  The permanganate solution changes from purple to colourless quickly.

Test for carboxylic acid group
Any one of the following:
- Add sodium hydrogencarbonate solution.
  Effervescence occurs.
- Test with moist blue litmus paper.
  The blue litmus paper turns red.

Part II Multiple choice questions

6 Propene reacts with bromine dissolved in an organic solvent to give

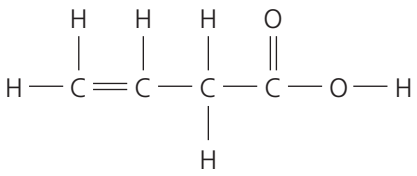
A 1-bromopropane.
B 2-bromopropane.
(C) 1,2-dibromopropane.
D 1-bromopropane and 2-bromopropane.

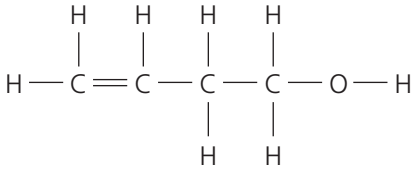
(HKCEE, Paper 2, 2011,13)

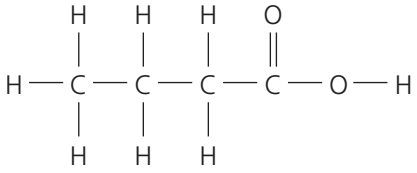
7 Which of these is a tertiary alcohol?
Refer to page T41 for explanation.
A 3-methylpentan-2-ol
B Pentan-2-ol
C Pentan-3-ol
(D) 2-methylpentan-2-ol

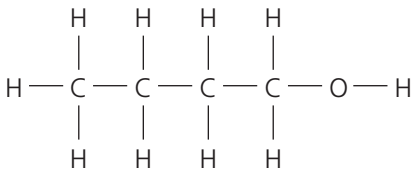
(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2009, 4)

8 Compound E rapidly decolourizes cold acidified potassium permanganate solution. When E is added to sodium hydrogencarbonate solution, effervescence occurs. Which of the following compounds may E be?

(A)


B


C


D


(HKCEE, Paper 2, 2010, 38)

9 The best method of converting ethanol, C_2H_5OH, into iodoethane, C_2H_5I, is to

A heat iodine and ethanol under reflux.
B react ethanol and potassium iodide in the presence of dilute acid.
C heat potassium iodide and ethanol with concentrated sulphuric acid.
(D) heat red phosphorus, ethanol and iodine under reflux.

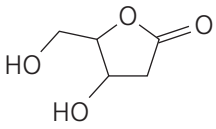
(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2010, 14)

10 What is the chemical formula of the organic product formed from the reaction between ethanol and propanoic acid under suitable conditions?

A $CH_3COOC_3H_7$
B $C_2H_5COOCH_3$
(C) $C_2H_5COOC_2H_5$
D $C_3H_7COOC_2H_5$

(HKCEE, Paper 2, 2011, 15)

11 The structure of an organic compound is shown below:



Which of the following statements is correct?

A The compound does NOT show enantiomerism.
B The molecular formula of the compound is $C_5H_6O_4$.
C The compound contains a ketone group.
(D) The compound can be oxidized by acidified $K_2Cr_2O_7$(aq).

(HKDSE, Paper 1A, 2012, 28)

12 When the following reaction mixtures are warmed, which will contain ethanoic acid as one of the products?
Refer to page T41 for explanation.
A Ethyl methanoate and sodium hydroxide solution
B Ethyl methanoate and dilute sulphuric acid
C Methyl ethanoate and sodium hydroxide solution
(D) Methyl ethanoate and dilute sulphuric acid

(Edexcel Advanced GCE, Unit 4, Jun. 2010, 9)

13 Consider the isomeric compounds shown below:

CH_3COCH_3 and CH_3CH_2CHO

Which of the following reagents can be used to distinguish between the two compounds?

(A) Acidified potassium dichromate solution
B Lithium aluminium hydride
C Dilute sulphuric acid
D pH indicator

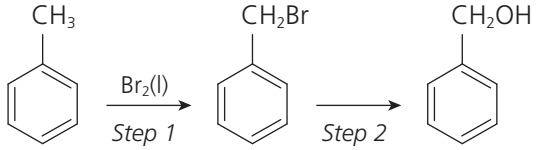
(HKDSE, Practice paper 1A, 2012, 27)

14 Which of the following statements concerning compound X ($CH_3CH=CHCH_2OH$) is / are correct?

(1) The systematic name of X is but-2-en-1-ol.
(2) X reacts with HBr to give a single product.
(3) X can exhibit *cis-trans* isomerism.

A (1) only
B (2) only
(C) (1) and (3) only
D (2) and (3) only

15 Consider the following conversion of organic compounds:



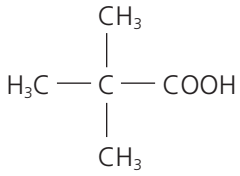
Which of the following statements about the above conversion are correct?

(1) Excess $Br_2(l)$ should be used in *Step 1*.
(2) Light is needed in *Step 1*.
(3) The reagent used in *Step 2* can be KOH(aq).

A (1) and (2) only
B (1) and (3) only
(C) (2) and (3) only
D (1), (2) and (3)

(HKDSE, Paper 1A, 2013, 34)

16 The structure of a compound is shown below:



Which of the following statements concerning the compound are correct?

(1) It can form a salt with aqueous ammonia.
(2) It can be reduced to an alkanol by using $LiAlH_4$.
(3) It can form an ester with methanol under suitable conditions.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

(HKDSE, Paper 1A, 2012, 34)

17 Which of the following compounds can be formed when $CH_3CH_2CH(OH)CH_2CH_3$ is dehydrated?
Refer to page T41 for explanation.

(1) $CH_2=CHCH_2CH_2CH_3$
(2) $CH_3CH=CHCH_2CH_3$
(3) $(CH_3)_2C=CHCH_3$

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

18 Which of the following statements about the action of sodium hydroxide solution on ethanamide is / are correct?

(1) Sodium ethanoate is formed in the reaction.
(2) In the reaction, sodium hydroxide acts as a catalyst.
(3) The reaction attains equilibrium if the reaction mixture is heated under reflux.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Paper 1A, 2013, 32)

14 (2) X reacts with HBr to give the products shown below.

$CH_3CH=CHCH_2OH \xrightarrow{HBr} CH_3CH_2CHBrCH_2OH + CH_3CHBrCH_2CH_2OH$

(3) $CH_3CH=CHCH_2OH$ has two different atoms or groups of atoms on each carbon atom of the carbon-carbon double bond. It can exhibit cis-trans isomerism.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

19 Give the structural formula(e) of the major product(s) expected from each of the following reactions.

Refer to page T42.

a) CH_2CH_2OH → ($Na_2Cr_2O_7 / H_3O^+$, heat)

b) H_3C–(benzene)–$COOH$ → (1 $LiAlH_4$ / ethoxyethane; 2 H_3O^+)

c) $CH_3CH_2COOCH_2CH_3$ → (NaOH(aq))

d) CH_3CONH_2 → (NaOH(aq))

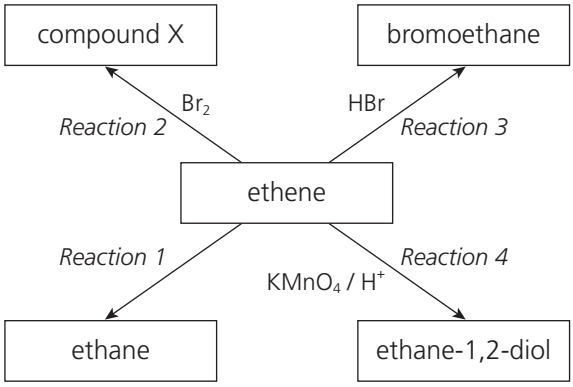
e) $CH{=}CHCH$ (benzene) → ($NaBH_4 / H_2O$)

f) CH_3 (naphthalene) → (Cl_2, UV light)

g) $OCCH_3$, $COOH$ (benzene) → (OH^-(aq), heat)

h) $CH_3CH_2CH_2OH$ → (excess conc. H_2SO_4, 180 °C)

20 a) Study the reaction scheme below and then answer the questions that follow.



i) Name the reagent and catalyst needed for *Reaction 1*.

Reagent: hydrogen
Catalyst: nickel / palladium / platinum

ii) Give the systematic name and structural formula of compound X.

iii) Describe what colour change you would see during *Reaction 4* if a small amount of acidified $KMnO_4$(aq) was shaken with ethene.

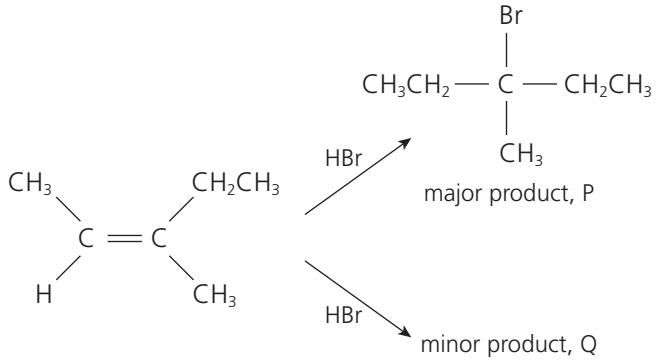
From purple to colourless

b) Explain why the alkene, propene, could form two products when it reacts with hydrogen bromide in a similar way.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2010, 18(b)–(c))

The bromine atom can add to either carbon atom of the carbon-carbon double bond, so forming either 1-bromopropane or 2-bromopropane.

21 The alkene 3-methylpent-2-ene reacts with hydrogen bromide as shown below.



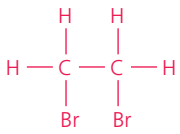
a) Name the major product P.

3-bromo-3-methylpentane

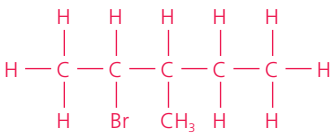
b) Draw the structure for the minor product Q and state the type of structural isomerism shown by P and Q.

(AQA Advanced Subsidiary, Unit 2, Jun. 2010, 6(a))

20 a) ii) 1,2-dibromoethane



21 b)



Position isomerism

Unit Exercise

22 This question is about the reactions and properties of some haloalkanes.

a) 1-bromobutane can be converted into butan-1-ol.

i) State the reagents and conditions needed for the conversion.
Sodium hydroxide solution, heat under reflux

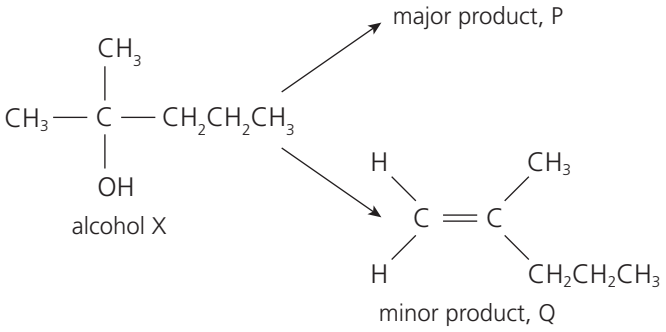
ii) Name the type of reaction involved.
Substitution

iii) Write a chemical equation for the reaction.

b) 1-chloropropane can be prepared by reacting propan-1-ol with potassium chloride in the presence of concentrated sulphuric acid.

Explain why a similar reaction using potassium iodide and concentrated sulphuric acid should NOT be used to prepare 1-iodopropane.
Concentrated sulphuric acid is a strong oxidizing agent. It oxidizes potassium iodide to iodine.

23 Alcohol X undergoes dehydration to give alkenes.

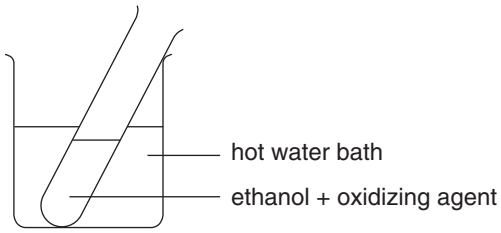


a) Give the IUPAC name of alcohol X.
2-methylpentan-2-ol

b) Draw the structure of the major product, P.

c) State the type of structural isomerism shown by P and Q.
Position isomerism

24 A student attempted to oxidize ethanol to ethanoic acid using the set-up shown below.



a) Suggest an oxidizing agent that can be used.

b) State one advantage of using a hot water bath over direct heating with a Bunsen burner in carrying out the experiment.

c) The student failed to obtain ethanoic acid even after a long period of time. The student then used Quickfit apparatus to perform the experiment. After some time, ethanoic acid was finally obtained.

i) Draw a labelled diagram to show how to set up Quickfit apparatus for carrying out the experiment.

ii) Explain why ethanoic acid could finally be obtained.

(HKCEE, Paper 1, 2009, 8(a)–(c))

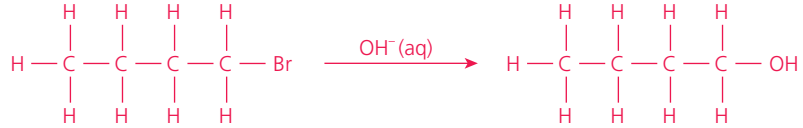
25 Ethanol can be used to make ethanal.
Refer to page T43.

a) Identify the two chemicals you would use to make ethanal from ethanol in the laboratory.

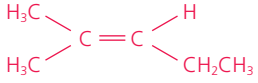
b) Draw a diagram of the apparatus you would use to prepare ethanal from ethanol in the laboratory and collect the product.

(Edexcel Advanced Subsidiary, Unit 2, Jan. 2011, 14(c))

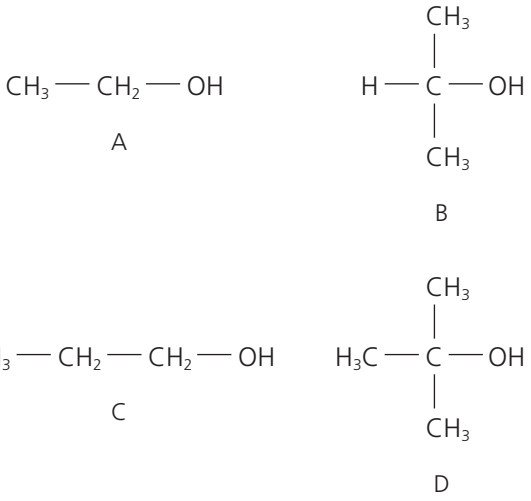
22 a) iii)



23 b)



26 Alcohols A, B, C and D are shown below.



a) A student heated each alcohol, A–D, with acidified potassium dichromate solution as the oxidizing agent. With alcohols A, B and C, the colour turned from orange to green.

Identify the organic product and write a balanced equation for the reaction of alcohol B with acidified potassium dichromate solution.

Use [O] to represent the oxidizing agent, acidified potassium dichromate solution.

b) The student heated alcohol D with ethanoic acid in the presence of an acid catalyst. An organic product E was formed with a fruity smell.

i) Name alcohol D.

ii) Name the functional group in the organic product E.

iii) Draw the structure of the organic product E.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2009, 5(b)–(c))

26 a) Organic product: $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$
Equation: $(CH_3)_2CHOH + [O] \longrightarrow (CH_3)_2CO + H_2O$

b) i) methylpropan-2-ol
ii) Ester functional group

iii) $H_3C-\overset{\overset{O}{\|}}{C}-O-C(CH_3)_3$

27 A student decided to make some propene from propan-1-ol.

Propan-1-ol is a liquid at room temperature, whilst propene is a gas. The strongest type of intermolecular attraction between propan-1-ol molecules is hydrogen bonding.

a) i) Draw a diagram to show the hydrogen bonding between two propan-1-ol molecules. Include relevant lone pairs and partial charges in your diagram. Refer to page T43.

ii) Propan-1-ol has a higher boiling point than propene. Explain this in terms of intermolecular attraction.

b) The equation below represents the reaction that occurs when propene forms from propan-1-ol.

$CH_3CH_2CH_2OH(g) \rightleftharpoons CH_3CH{=}CH_2(g) + H_2O(g)$

i) Underline the term below that describes the type of reaction occurring in the above equation.

addition elimination hydrolysis substitution

ii) Give the reagents and conditions required for the reaction represented by the above equation.

c) The student used the propene for a range of chemical reactions. Give the structural formula of a product that forms if the student reacts the propene with:
Refer to page T43.
i) hydrogen bromide

ii) hydrogen

d) The student carried out the reaction of propene with hydrogen at room temperature and pressure by carefully choosing the catalyst to use. Name the catalyst that would be required under these reaction conditions. Platinum

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2010, 4(a)–(c), (g)–(h))

27 a) ii) The boiling point of a carbon compound depends on the strength of its intermolecular attractions.
The intermolecular attractions in propan-1-ol are mainly hydrogen bonds while those in propene are van der Waals' forces.
Hydrogen bonds are stronger than van der Waals' forces.
Thus, propan-1-ol has a higher boiling point than propene.

b) ii) Any one of the following:
- Concentrated sulphuric acid / phosphoric acid
  Heat / reflux
- Aluminium oxide / silica / pumice
  Heat, propan-1-ol vapour passes over the catalyst

28 This question is about the six alcohols below.

Refer to page T44.

butan-2-ol	ethane-1,2-diol
2-methylpentan-3-ol	methylpropan-2-ol
propan-1-ol	propan-2-ol

a) Which alcohol is an example of a tertiary alcohol?

b) Draw the skeletal formula for 2-methylpentan-3-ol.

c) Butan-2-ol and methylpropan-2-ol are structural isomers.

i) What is meant by the term 'structural isomer'?

ii) Draw another structural isomer of these two alcohols.

d) Ethane-1,2-diol can be dissolved in water to act as an anti-freeze in car radiators.

Explain why ethane-1,2-diol is very soluble in water.

e) Ethane-1,2-diol is heated under reflux with ethanoic acid and a small amount of H_2SO_4 catalyst. Compound A is formed with molecular formula $C_6H_{10}O_4$.

Draw the structure of compound A.

f) Butan-2-ol is heated with H_2SO_4 catalyst.

- A mixture of three alkenes (B, C and D) forms.
- The alkenes B and C are stereoisomers.

i) Draw the structures of the two stereoisomers B and C.

ii) What type of stereoisomerism is shown by B and C?

iii) Draw the structure of the other alkene, D, that is formed in this reaction.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jan. 2010, 6(a)–(f))

31 Reagent for reduction — lithium tetrahydridoaluminate in anhydrous ethoxyethane, followed by hydrolysis using dilute acid or sodium tetrahydridoborate in water

$CH_3COCHO + 4[H] \longrightarrow CH_3CH(OH)CH_2OH$

Reagent for oxidation — acidified potassium dichromate solution, the dichromate solution changes from orange to green in the process

$CH_3COCHO + [O] \longrightarrow CH_3COCOOH$

29 Compound X has the structural formula shown below:

Refer to page T44.

$(CH_3)_2C(OH)CH_2CH_3$

Compound X can be dehydrated by passing its vapour over a heated solid W.

a) Name solid W.

b) Draw a labelled diagram of the set-up you would use for the reaction.

c) Draw the structures of two alkenes formed in the reaction.

d) Suggest whether the two alkenes formed can exhibit *cis-trans* isomerism. Explain.

e) Compound Y shown below is a structural isomer of compound X.

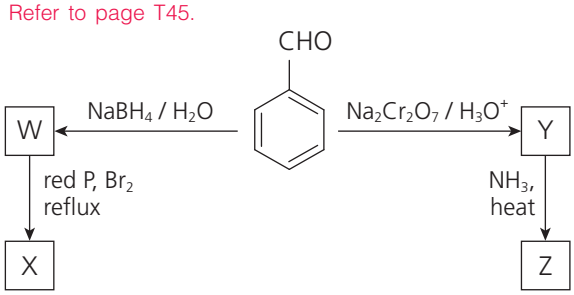
$CH_3CH_2CH(OH)CH_2CH_3$

i) State the type of structural isomerism involved.

ii) Suggest a chemical test to distinguish between X and Y.

30 Benzaldehyde is a starting material for many kinds of organic transformations. For the reaction scheme involving benzaldehyde shown below, give the structures of the organic products W, X, Y and Z.

Refer to page T45.



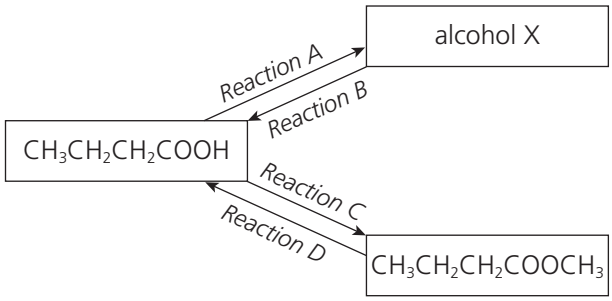
31 'Methylglyoxal', CH_3COCHO, is formed in the body during metabolism.

Describe ONE reduction reaction and ONE oxidation reaction of methylglyoxal that could be carried out in the laboratory.

Your answer should include reagents, equations and observations, if any.

(OCR Advanced GCE, Chem. A, F324, Jun. 2011, 4)

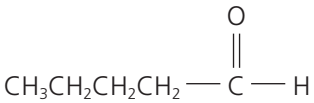
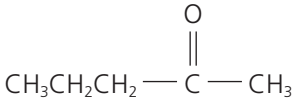
32 Some reactions involving butanoic acid are shown below.



a) Draw the structure of alcohol X.
$CH_3CH_2CH_2CH_2OH$

b) Suggest the reactant(s) and conditions for each of *Reactions A*, *B*, *C* and *D*.
Refer to page T45.

33 The table below shows the structures of three compounds with the molecular formula $C_5H_{10}O$.

Compound X $H_3CCH{=}CHCH(OH)CH_3$	pent-3-en-2-ol
Compound Y 	pentanal
Compound Z 	pentan-2-one

a) Complete the table by giving the IUPAC names of compounds X and Z.

b) State the type of structural isomerism shown by these three compounds.
Functional group isomerism

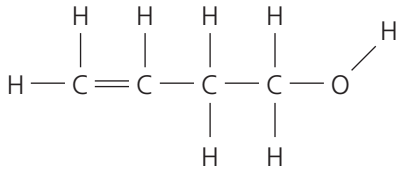
c) Compound X exists in two isomeric forms.
Refer to page T45.
i) Draw the structure of each isomer.

ii) State the type of isomerism involved.

iii) Identify the feature of the double bond in compound X that causes the existence of two isomeric forms.

d) Suggest a chemical test for distinguishing between compounds Y and Z. State also your expected observation.
Warm each compound with acidified potassium dichromate solution.
Compound Y turns the dichromate solution from orange to green.
There is no observable change for compound Z.

34 An organic compound A has the structure shown below.
Refer to page T46.



a) Give the observations that you would expect to make when each of the tests below is carried out. Give a brief chemical explanation of each reaction that occurs.

i) A small amount of phosphorus pentachloride is added to 2 cm^3 of A in a test tube.

ii) A few drops of potassium permanganate solution and 2 cm^3 of dilute sulphuric acid are added to 2 cm^3 of A in a test tube and the mixture is gently warmed.

iii) A few drops of aqueous bromine are added to 2 cm^3 of A in a test tube and the mixture is shaken.

b) Draw the structure of the product formed in (a)(iii).

(Edexcel Advanced Subsidiary GCE, Unit 3B, May 2009, 3(a)–(b))

35 Compound X ($C_9H_8O_2$) is a mono-substituted aromatic compound. Deduce the possible structures of compounds X and Y based on the information below:

(1) Compound X exhibits *cis-trans* isomerism.

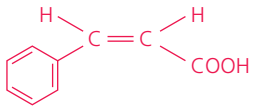
(2) Compound X reacts with $H_2(g)$, in the presence of a catalyst, to give compound Y.

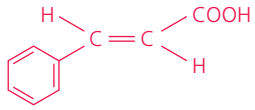
(3) Effervescence occurs when compound Y is treated with $NaHCO_3(aq)$.

From (3), Y contains –COOH group. This suggests that X also contains –COOH group.

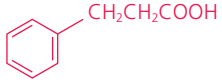
From (1) and (2), X contains C=C group, and there are two different atoms or groups of atoms on each carbon atom of the C=C group.

∴ possible structures of X



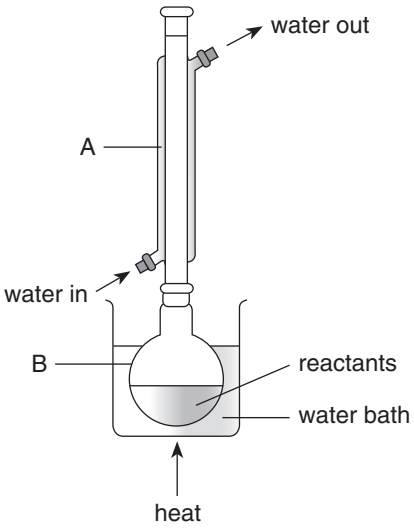


possible structure of Y



36 Fruit flavourings often contain esters. The reaction used to make an ester is very slow.

This apparatus can be used to make an ester.

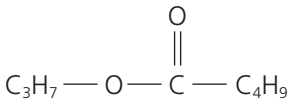


a) i) Why is it necessary to heat the reactants in flask B?
To speed up the reaction / reduce the time taken.

ii) What is the purpose of the apparatus labelled A in the diagram? To condense vapour and prevent the loss of volatile substances.

b) i) Propyl pentanoate is an ester which is used in pineapple flavouring. It is made from the reaction of an alcohol with a carboxylic acid.

A molecule of propyl pentanoate can be represented as below.



Which TWO compounds would be used to make propyl pentanoate? Choose from the list below.

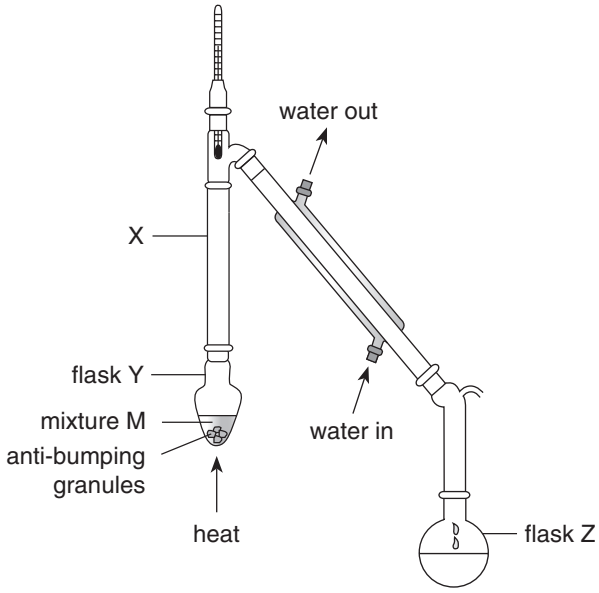
C_3H_7OH	C_4H_9OH	C_3H_8
C_3H_7COOH	C_4H_9COOH	C_4H_{10}

ii) An inorganic acid is used as a catalyst in the reaction.

Write down the name of the catalyst which is used. Sulphuric acid

(OCR GCSE 21st Century Science (Higher Tier), Additional Applied Science A, A335, Jan. 2012, 3)

37 In a certain experiment, a mixture of methyl propanoate and sodium hydroxide solution was heated under reflux for some time. The resulting mixture M was then transferred to flask Y and heated as shown below.



a) i) Write a chemical equation for the reaction between methyl propanoate and sodium hydroxide solution.
Refer to page T46.

ii) Name the type of reaction taking place.
Hydrolysis

b) i) Name apparatus X.
Air condenser

b) ii) Serve as a fractionating column for separating methanol from the products formed.

ii) What was the function of apparatus X?

c) Name the distillate collected in flask Z.
Methanol

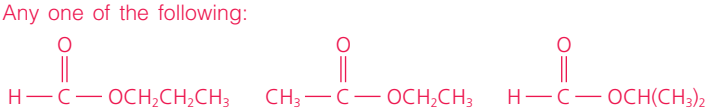
d) What was the purpose of using anti-bumping granules?
To ensure even boiling.

e) Which one of the following hazard warning labels should be displayed on a bottle of methyl propanoate? B



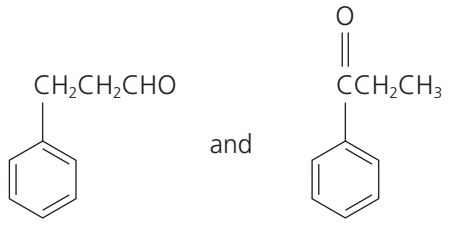
f) Draw the structure of another ester which has the same molecular formula as methyl propanoate, and give its systematic name.

Any one of the following:



Unit Exercise

38 Describe, by giving reagent(s) and stating observations, how you could distinguish between compounds in each pair using a simple test tube reaction.
Refer to page T46.

a) 

b) 

and $CH_2{=}CHCH_2OH$

c) CH_3CH_2COOH and CH_3COCH_2OH

39 Acyclic carbon compound X contains carbon, hydrogen and oxygen only. The relative molecular mass of X is 58.0. Analysis of X gives the following results:

Test 1 Complete combustion of 1.45 g of X gives 3.30 g of carbon dioxide and 1.35 g of water.

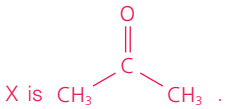
Test 2 X gives a negative result when warmed with acidified $K_2Cr_2O_7$(aq).

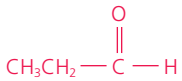
a) Deduce the molecular formula of X.
C_3H_6O; refer to page T47.
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

b) Draw ONE possible structure of X.

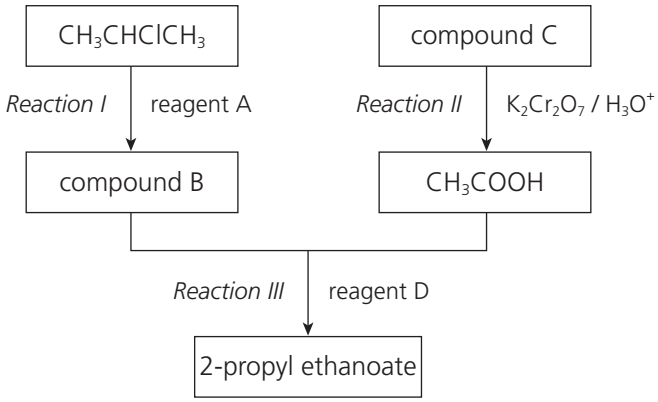
c) Compound Y is a structural isomer of X. Draw ONE possible structure of Y.

39 b) X gives a negative result when warmed with acidified $K_2Cr_2O_7$(aq), so X is not aldehyde, and primary and secondary alcohol.



c) 

40 The ester 2-propyl ethanoate is a solvent used in perfumes. A chemist uses the following reaction scheme for the synthesis of the ester.



a) Identify reagent A.
Sodium hydroxide solution

b) Compound B is an alcohol.

i) Write the structural formula of compound B.

ii) Give the IUPAC name of compound B.
propan-2-ol

c) Compound C is also an alcohol.

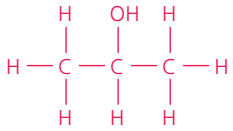
i) Give the IUPAC name of compound C.
ethanol

ii) Suggest TWO measures the chemist may take to ensure the maximum possible yield of CH_3COOH is obtained. Heat under reflux.
Use excess oxidizing agent / $K_2Cr_2O_7$.

d) Identify reagent D.
Concentrated sulphuric acid

e) Write the structural formula of 2-propyl ethanoate.

f) Which of the reactions in the reaction scheme is an oxidation reaction?

40 b) i) 

e) 

f) *Reaction II*

Unit Exercise